08. Physically Based Animations

Aka. Simulations

INFO-H502

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LISA-VR

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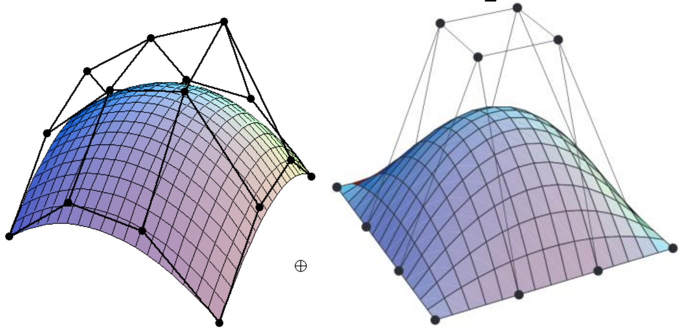
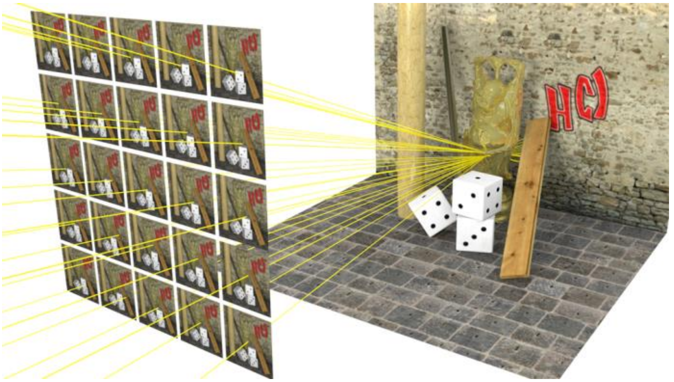
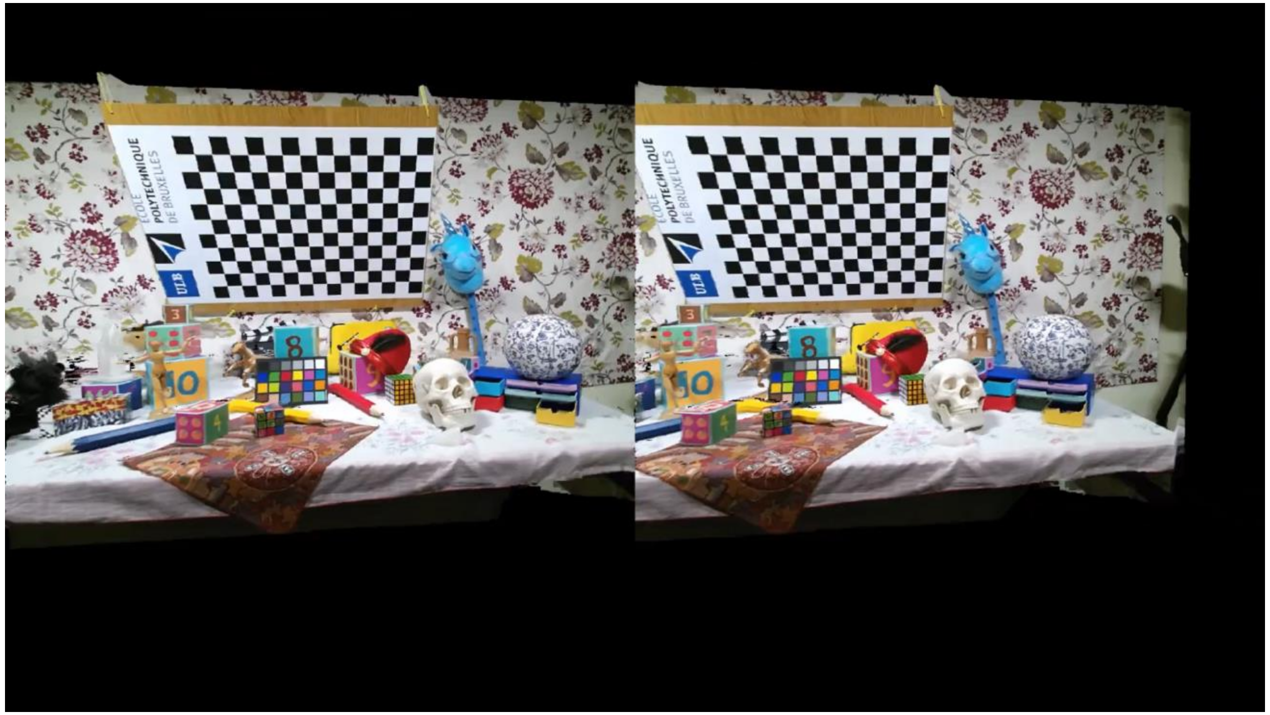
Previously



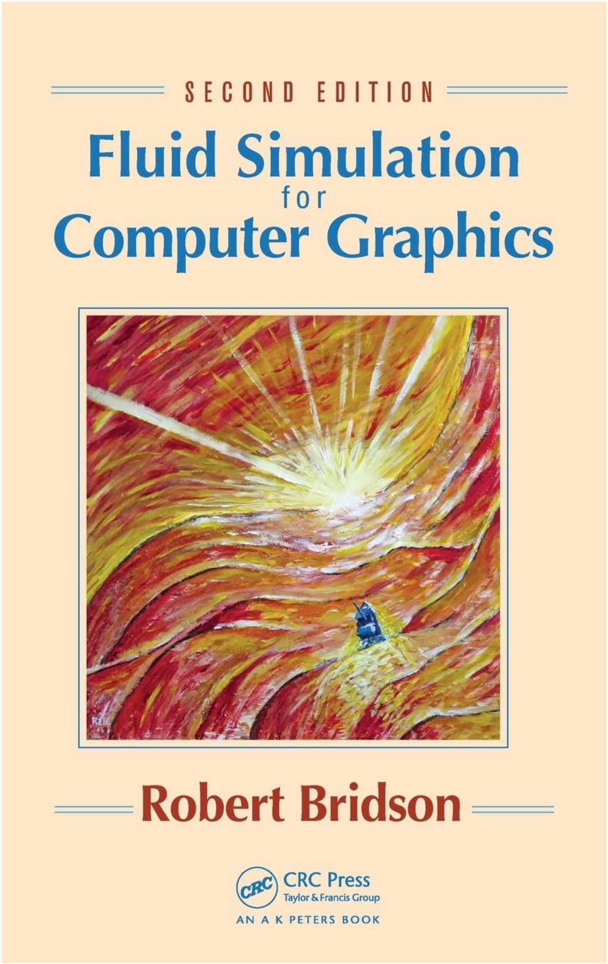




Bibliography

- Those lectures were heavily inspired by Batty notes.
- He maintains a website on the state of the art of physically based animations
  - https://www.physicsbasedanimation.com/

- And this book for fluids:
  - If you are interested in fluids, you should really get a copy
  - Its a phenomenal book



Why Physically Based Animations?

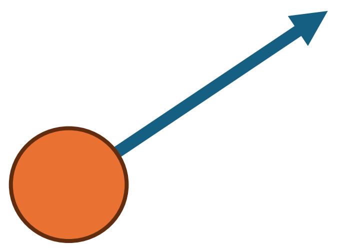
- Physical based animations allows us to simulate Physics !
- They use the laws of motion, gravity, and other forces

- We can for example simulate:
  - Hairs
  - Fluids
  - Particle systems (fire, planets)
  - Partial differential equations

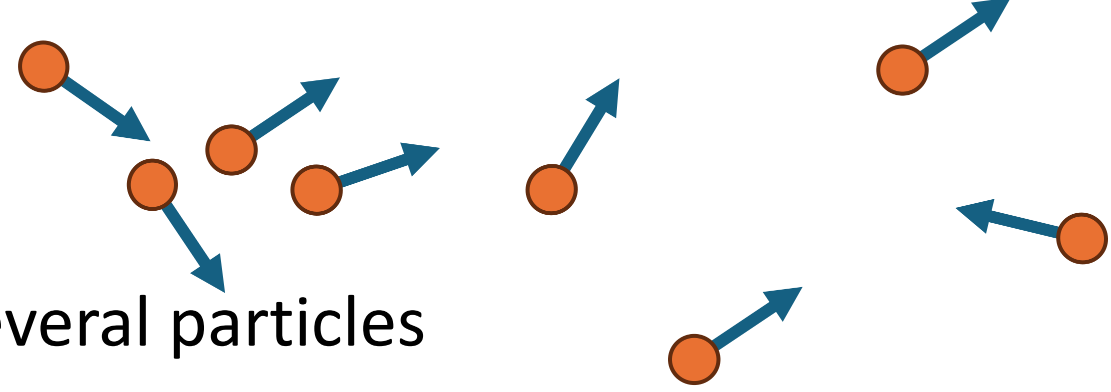
- This lecture will summarize a lot of material you have seen in:
  - Mechanics 2
  - Numerical Analysis
  - Fluid mechanics

A Particle – The Simplest Element

- In most physical simulations, we want to move objects
- The simplest object we can animate is a **particle**
- A particle has a position $\vec{x}$ and a velocity $\vec{u}$
  - It can also have more attributes, such as a mass, temperature, etc.



Particle Systems



- Having only one particle is not that useful
- To make for example fireworks, we need several particles
- **Particle systems** are a set of particles

- They are usually implemented as a class responsible for the animation, for example a fire contains a lot of particles, each contains position, velocity, luminosity, color, end of life, orientation, mass, temperature, etc..

- At each iteration, we update the position of each particle in the system.

- Simple system have particles that do not interact among each others

Particle Systems – Lifecycle

- At each frame:
  - Create new particles and assign initial values
  - Update existing particle position/velocity/attributes according to a rule
  - Delete expired particles

- Rules can incorporate randomness

- Designing rules = Art

How Does A Particle Moves ?

- Newton Laws of Motion
- $x(t)$ Position of a particle in time
- $v = \frac{dx}{dt} = \dot{x}$ Speed
- $a = \frac{dv}{dt} = \frac{d^2x}{dt^2} = \ddot{x}$ Acceleration
- $p = mv$ Linear momentum
- Newton 2nd law: $F = ma$

- A very useful notation is to create a vector $A(t) = \begin{pmatrix} x(t) \\ v(t) \end{pmatrix}$
- So that we have $\frac{d}{dt}A(t) = \begin{pmatrix} v(t) \\ a(t) \end{pmatrix} = \begin{pmatrix} v(t) \\ F(t)/m \end{pmatrix}$

How Does A Particle Move ?

- Changing position or velocity is bad in computer graphics
  - It leads to jumps that are not visually pleasant
  - We only set the initial position and velocity
- Instead, we change the acceleration
  - Leads to smooth animations, but requires to find the velocity and position

- To get $v(t)$, integrate $a(t)$
  - $v(t) = \int a(t)dt = at + v_0$
- To get $x(t)$, integrate $v(t)$
  - $x(t) = \int v(t)dt = \frac{1}{2}at^2 + v_0 t + x_0$

- But, how to evaluate those integrals in a computer?

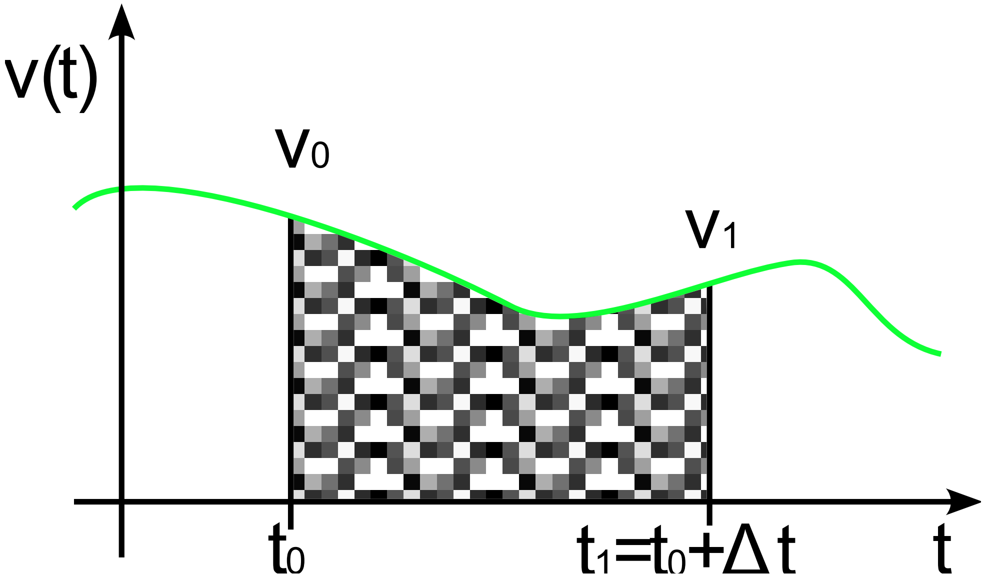
But How To Evaluate ?

- We have first order differential equations, i.e: $\dot{v} = a$ and $\dot{x} = v$
- To obtain the position x and the velocity v we want to integrate
- If the expression of v or a is simple, we can obtain an analytic solution.
- But in general, we can't ! The acceleration will depends on many forces with multiple interdependencies on the parameters.
- Instead, let's evaluate it numerically:
- First try – discretizing the derivative:
- $\frac{dx}{dt} = v \Rightarrow \frac{x_{t+1} - x_t}{\Delta t} = v(t)$

But How To Evaluate ?

- We have first order differential equations, i.e: $\dot{v} = a$ and $\dot{x} = v$

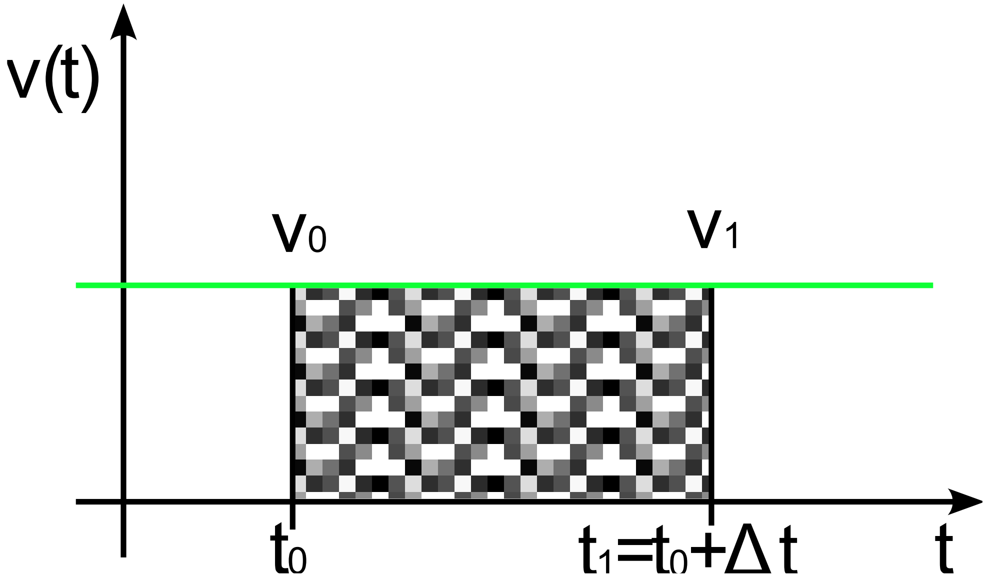
- To obtain the velocity v (or the position x) we want to integrate
  - i.e, to obtain the area under the curve



But How To Evaluate ?

- We have first order differential equations, i.e: $\dot{v} = a$ and $\dot{x} = v$

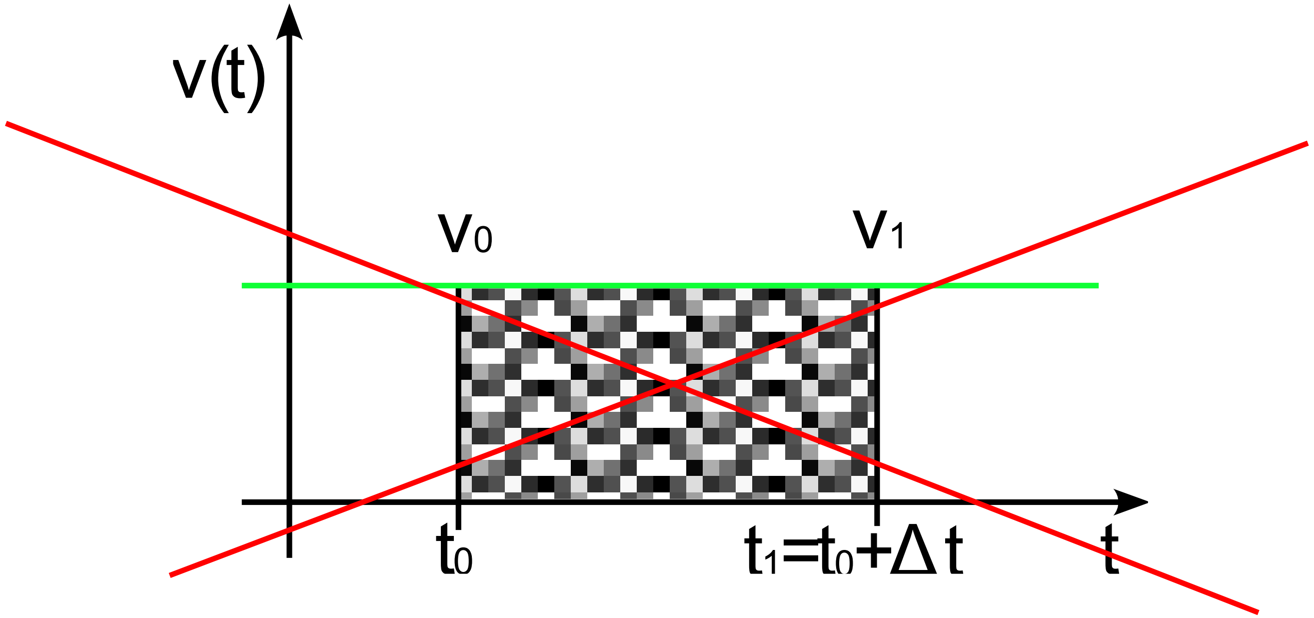
- To obtain the velocity v (or the position x) we want to integrate
- If the expression of v (or a) is constant, its easy.
  - $\int_a^b \dot{x} dt = \int_a^b v dt \Leftrightarrow x = v[1]_a^b \Leftrightarrow x = v(b - a)$



But How To Evaluate ?

- We have first order differential equations, i.e: $\dot{v} = a$ and $\dot{x} = v$

- To obtain the velocity v (or the position x) we want to integrate
- If the expression of v (or a) is constant, its easy.
  - But in general.. It's not.. The acceleration a depend on a (complex) force ($F = ma$)

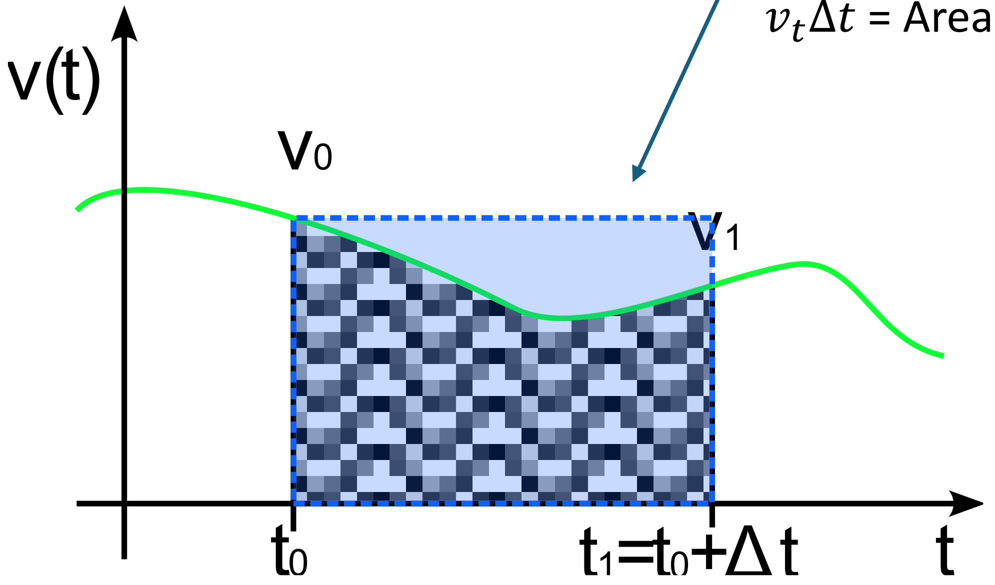


But How To Evaluate ?

- We have first order differential equations, i.e: $\dot{v} = a$ and $\dot{x} = v$

- Instead, we will evaluate this integral using numerical methods !
- First attempt of discretization:
- $\frac{dx}{dt} = v(t) \Rightarrow \frac{x_{t+\Delta t} - x_t}{\Delta t} = v_t \Leftrightarrow x_{t+\Delta t} - x_t = v_t \Delta \mathrm{t}$

$v_t \Delta t$ = Area of the box



Forward Euler

- Because $x_{t+\Delta t} - x_t = v_\mathrm{t}\Delta\mathrm{t}$ corresponds to the area of a box with height v_t and width Δt, we call this method Forward Euler

- Rewriting our equations:
  - $x_{t+\Delta t} = x_t + v_t \Delta t$
  - $v_{t+\Delta t} = v_t + a_t \Delta t$

- Is it accurate ?

Forward Euler

- Forward Euler:
  - $x_{t+\Delta t} = x_t + v_t \Delta t$
  - $v_{t+\Delta t} = v_t + a_t \Delta t$

- Is it accurate ?

- Simple example: $v(t) = \frac{dx}{dt} = -x,\ \ x(0) = 1$
- Analytically this differential equation has solution:
  - $\frac{\mathrm{dx}}{-x} = \mathrm{dt} \Rightarrow \int \frac{dx}{x} = \int -1\, dt \Leftrightarrow \ln(x) = -t + c \Leftrightarrow x(t) = e^{-t+k} \Leftrightarrow x(t) = ce^{-t}$
  - As $x(0) = 1$, we have $x(0) = ce^{-0} = 1 \Leftrightarrow c = 1 \Rightarrow x(t) = e^{-t}$

	t	0	1	2	3	4	5	6	7	8
	x analytic	1	$3{,}6e^{-1}$	$1{,}3.10^{-1}$	5.10^{-2}	2.10^{-2}	7.10^{-3}	3.10^{-3}	9.10^{-4}	3.10^{-4}
$\Delta t = 11$	x Forward Euler	1	-1.10^{1}	1.10^{2}	-1.10^{3}	1.10^{4}	-1.10^{5}	1.10^{6}	-1.10^{7}	1.10^{8}

Forward Euler

Forward Euler is not monotonic !

- Forward Euler:
  - $x_{t+\Delta t} = x_t + v_t \Delta t$
  - $v_{t+\Delta t} = v_t + a_t \Delta t$

- Is it accurate ?

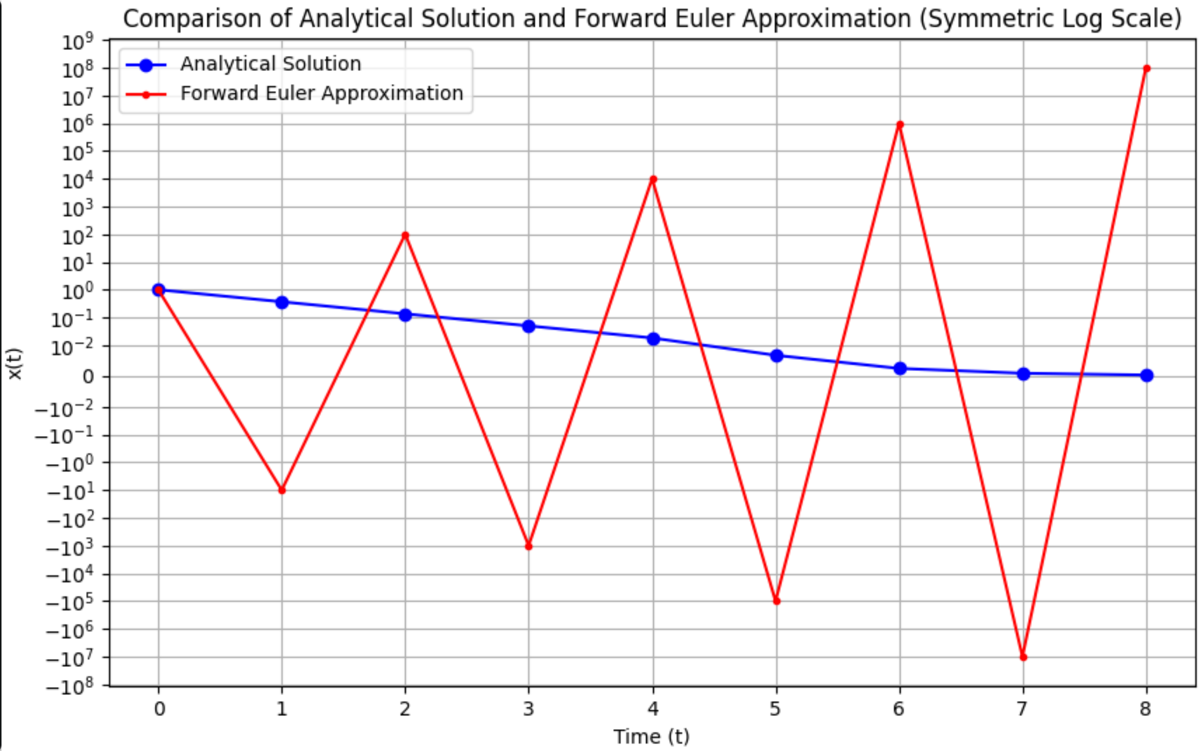
```python
import math

def print_scientific(numbers):
    for x in numbers:
        print(f"{x:.2e}", end=" ")
    print()

def analytic(n):
    x = []
    for t in range(n+1):
        x.append(math.exp(-t))
    return x

def forward_euler(n, Dt, x0):
    x = [x0]
    for t in range(1, n+1):
        x.append(x[t-1] + (-x[t-1]) * Dt)
    return x

print_scientific(analytic(8))
print_scientific(forward_euler(8, 11, 1))
```

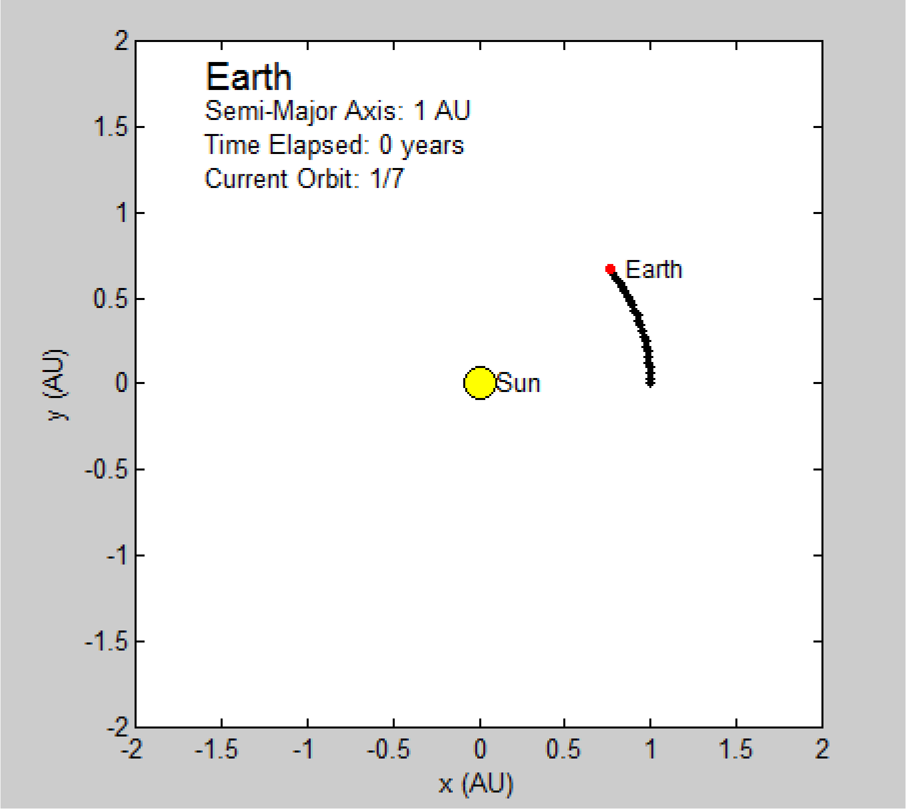


- Simple example: $v(t) = \dfrac{dx}{dt} = -x, \;\; x(0) = 1$

	t	0	1	2	3	4	5	6	7	8
	x analytic	1	$3{,}6e^{-1}$	$1{,}3.10^{-1}$	5.10^{-2}	2.10^{-2}	7.10^{-3}	3.10^{-3}	9.10^{-4}	3.10^{-4}
$\Delta t = 11$	x Forward Euler	1	-1.10^{1}	1.10^{2}	-1.10^{3}	1.10^{4}	-1.10^{5}	1.10^{6}	-1.10^{7}	1.10^{8}

Forward Euler

- Forward Euler: $x_{t+\Delta t} = x_t + v_t \Delta t$

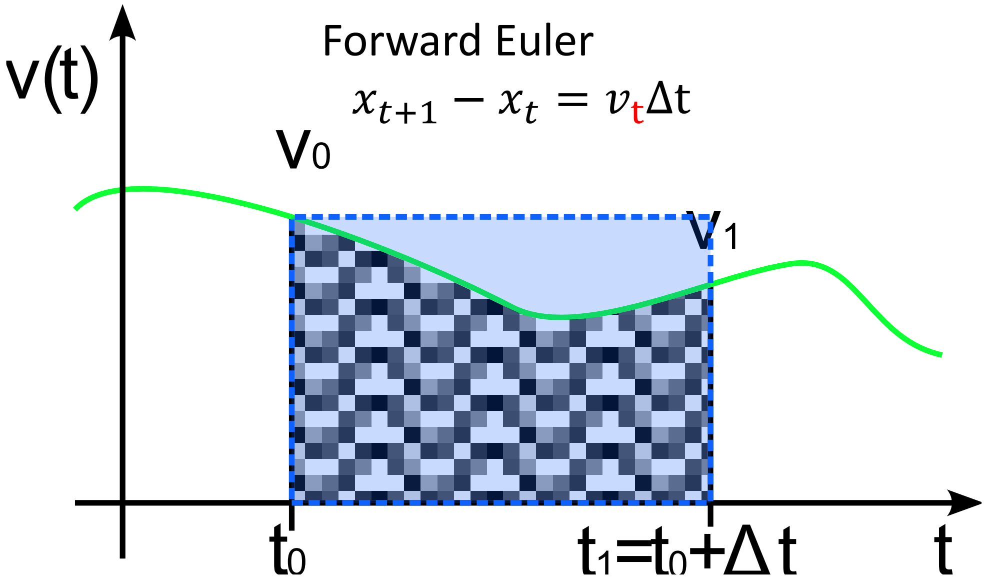


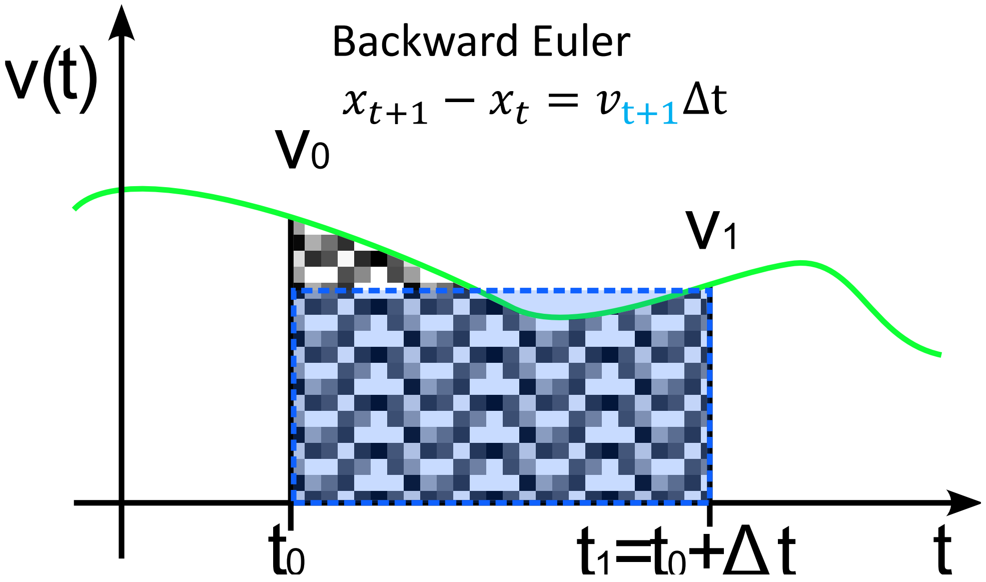
- Pros: Simple and fast
- Cons: Not accurate for large time steps, tends to be unstable

- We could increase the timesteps, but this is computationally expensive !

Backward Euler

- Instead of using the value of v at time t, we can use the value of v at time $t + \Delta t$
- This is backward Euler, it estimates the area starting from $v_{t+\Delta t}$
- $x_{t+\Delta t} - x_t = v_{t+\Delta t}\Delta t$
- But.. This requires to solve an equation to find $v_{t+\Delta t}$..





Backward Euler

- $x_{t+\Delta t} - x_t = v_{t+\Delta t}\Delta\mathrm{t}$
- But.. This requires to solve an equation to find $v_{t+\Delta t}$..
- Which may depend on $x_{t+\Delta t}$
- In our example, $v(t) = -x$, we need to evaluate 2 equations:
- $v(t + \Delta t) = -x_{t+\Delta t}$
- $x_{t+\Delta t} = x_t + v_{t+\Delta t}\Delta\mathrm{t} \Rightarrow x_{t+\Delta t} = x_t - x_{t+\Delta t}\Delta\mathrm{t}$

- Factoring $x_{t+\Delta t}$: $x_{t+\Delta t}(1 + \Delta t) = x_t \Leftrightarrow x_{t+\Delta t} = \frac{x_t}{1+\Delta t}$

Backward Euler

```python
import math

def print_scientific(numbers):
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    print()

def analytic(n):
    x = []
    for t in range(n+1):
        x.append(math.exp(-t))
    return x

def backward_euler(n, Dt, x0):
    x = [x0]
    for t in range(1, n+1):
        x_next = x[t-1] / (1 + Dt)
        x.append(x_next)
    return x

print_scientific(analytic(8))
print_scientific(forward_euler(8, 11, 1))
```
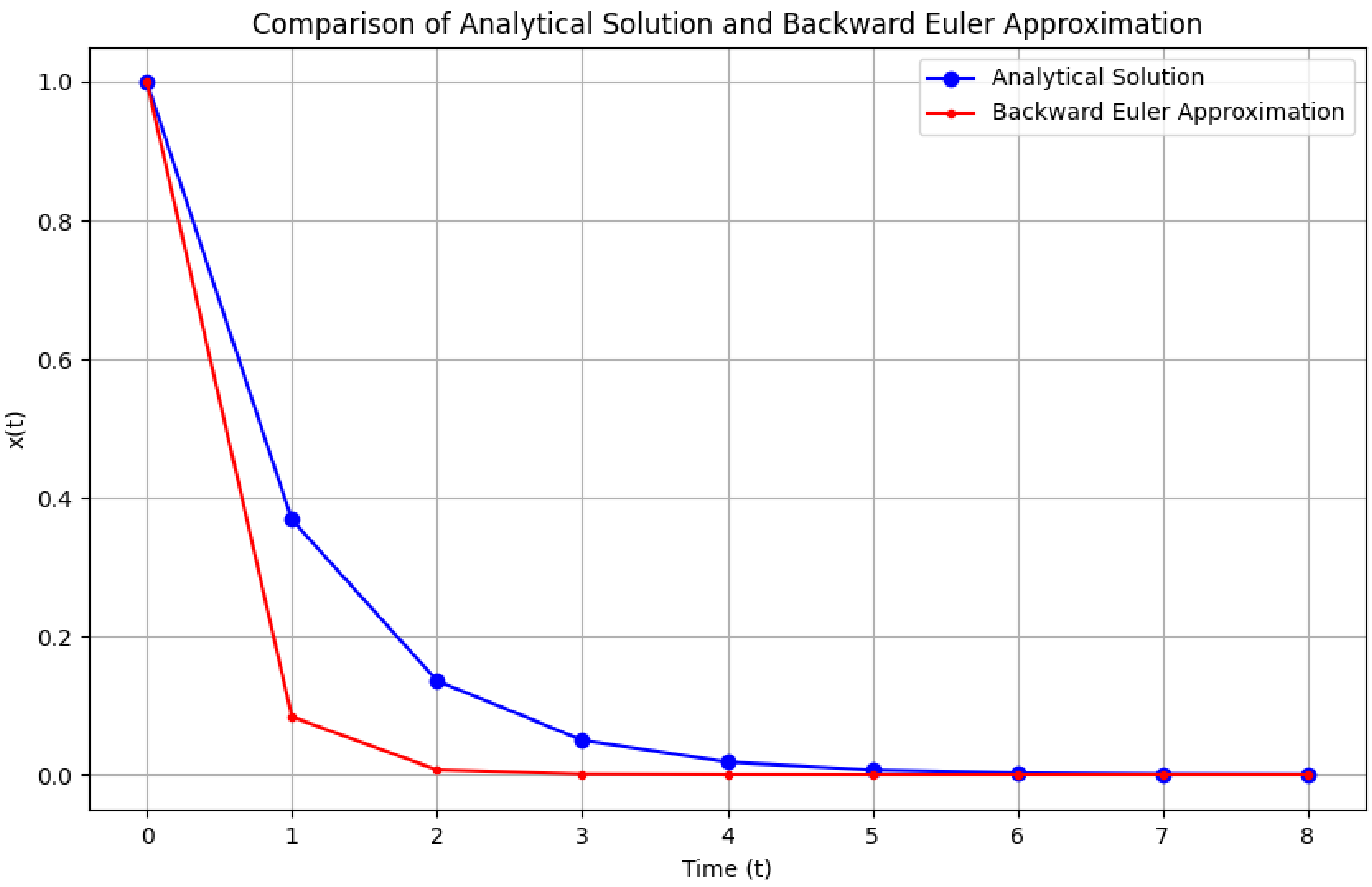



- It’s better, but we are not there yet, it overshoot quite a bit

Backward Euler

- Backward Euler: $x_{t+\Delta t} = x_t + v_{t+\Delta t}\Delta t$

- Pros: More stable for stiff systems and large time steps
- Cons: Requires solving equations at each time step, which can be computationally expensive

- What if we do not want to solve the equation for $v_{t+\Delta t}$ as it is expensive ?

Semi-Implicit (Symplectic) Euler

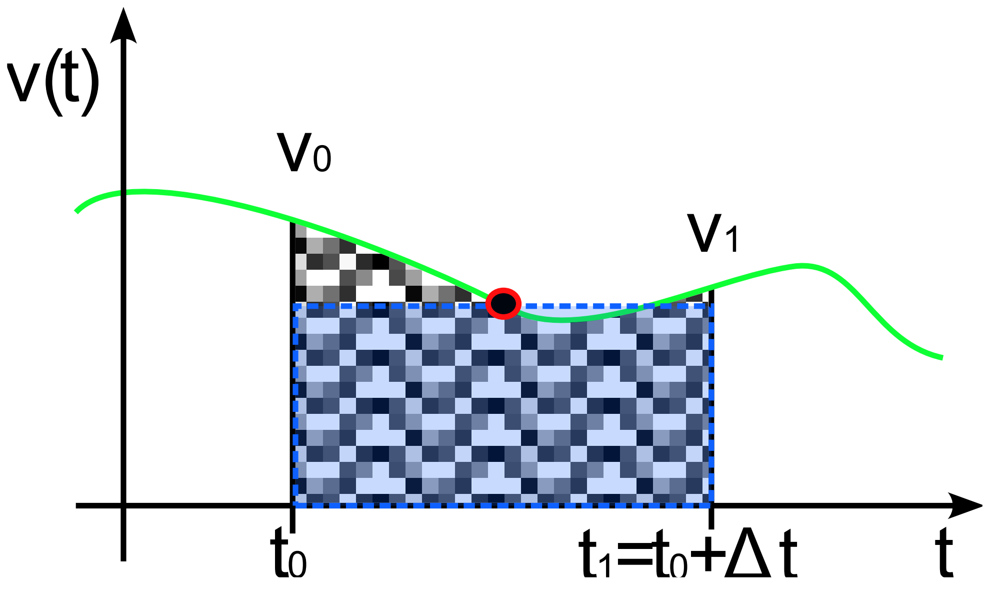
- Backward Euler: $x_{t+\Delta t} = x_t + v_{t+\Delta t}\Delta t$
- Instead of evaluating $v_{t+\Delta t}$, we can estimate it using acceleration and the current position !
- $v_{t+\Delta t} = v_t + a_t\Delta t$
- $x_{t+\Delta t} = x_t + v_{t+\Delta t}\Delta t$

- Pros: Better energy conservation properties, suitable for simulations involving oscillations (e.g., springs, pendulums), with appropriate Δt.
- Cons: Stability is limited for very stiff systems.

- Can we do better?

The Midpoint Method

- The Midpoint Method combine both Forward and Backward Euler by estimating the value of v at the midpoint of the interval.
- $v_{t+\frac{\Delta t}{2}} = v_t + a_t \frac{\Delta t}{2}$
- $x_{t+\Delta t} = x_t + v_{t+\frac{\Delta t}{2}} \Delta t$



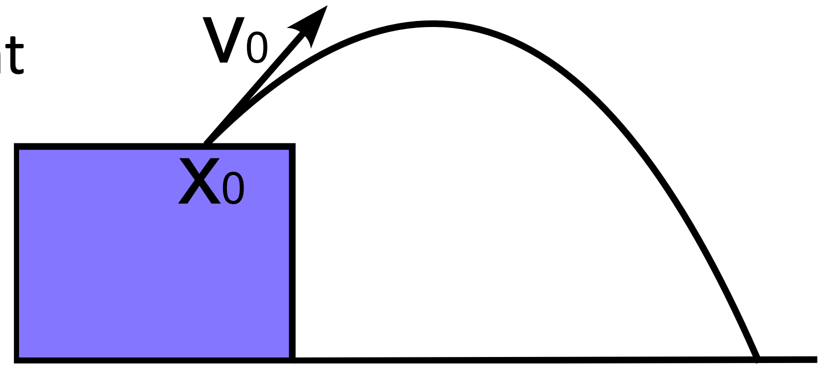
- Pros:
  - More accurate than Forward/Backward Euler for the same time step Δt.
  - Simple implementation while reducing error in the approximation.
- Cons:
  - Still requires knowledge of acceleration, similar to Backward Euler.

Multiple Forces

- Let's focus on the forces that define the acceleration
- $\sum F_i = ma$
- Example of forces:
  - Gravity
  - Wind (vector fields)
  - Springs (between particles)/ Elasticity
  - Damping / Viscosity
  - Friction
  - Collision/contact
  - Magnetism
  - User inputs
  - …
- We will see some of them

Gravity - Projectile Motion

- A first example of force: Gravity on a projectile
- We identify a projectile as a unique particle
  - We do not care about its rotation, only general movement
- Gravity force: $F_G = G\frac{m_1 m_2}{r^2}$
- Simplifying assumption on earth:
  - $m_1 =$ is constant (bigger than m_2)
  - r^2 is constant, earth diameter is big
- leading to $F_g = mg$ where $g = G\frac{m_1}{r^2} = 9{,}81 m/s^2$



Gravity - Projectile Motion

- In vector form: $a = (0, 0, -g)^T$
- Projectile position: $x(t) = \begin{pmatrix} x_x + v_x t \\ x_y + v_y t \\ x_z + v_z t + \frac{1}{2} g t^2 \end{pmatrix}$
  - Acceleration only affects 1_z: $x_x, x_y, x_z, v_x, v_y, v_z$ initial positions and velocities

- When does the projectile reach the maximum height ?
- $\Rightarrow$ When the derivative is zero:
  - $z'(t) = v_z + gt = 0 \Leftrightarrow t = -\frac{v_z}{g}$
- Pluging back into $z(t)$ to find the position:
  - $z(t) = x_z + v_z\left(-\frac{v_z}{g}\right) + \frac{1}{2} g \left(-\frac{v_z}{g}\right)^2 = x_z - \frac{v_z^2}{g} + \frac{1}{2}\frac{v_z^2}{g} = x_z + \frac{1}{2}\frac{v_z^2}{g}$

Gravity - Projectile Motion

- In this example, we integrated analytically
- We could do it because there was only a gravitational force
- Not very difficult to use physical equations.

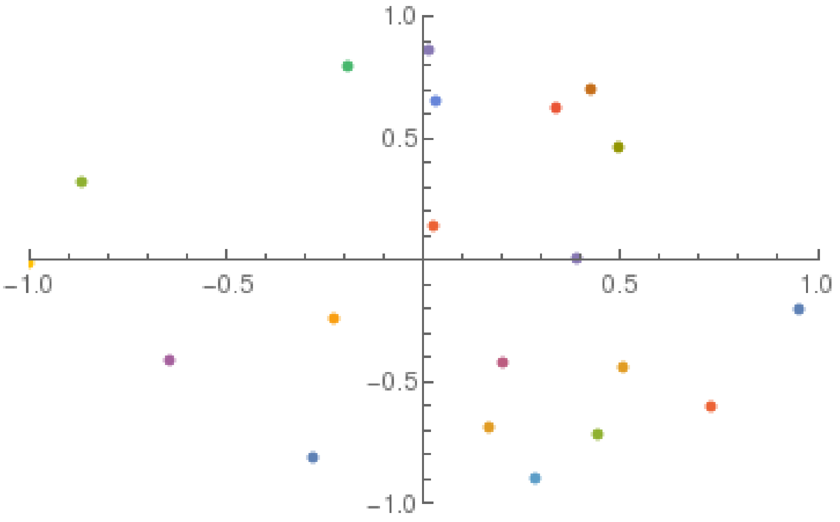
- But, as soon as more forces are involved, the integration is not easy to solve analytically !

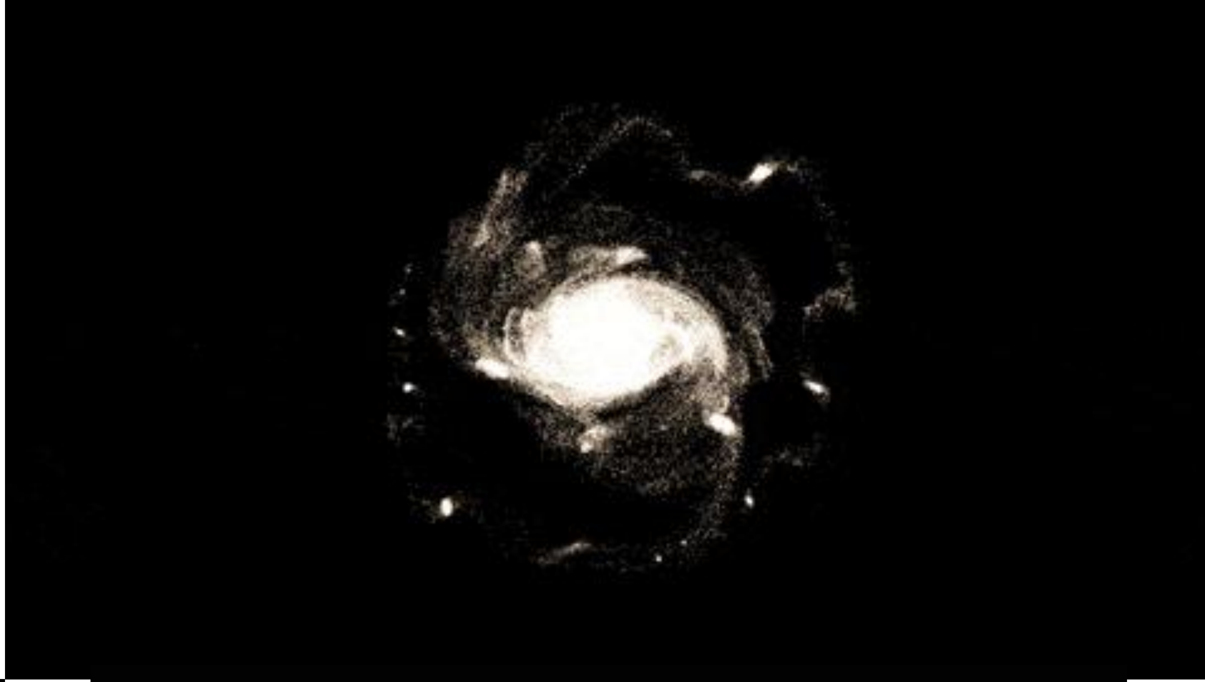
- We use approximations methods instead.

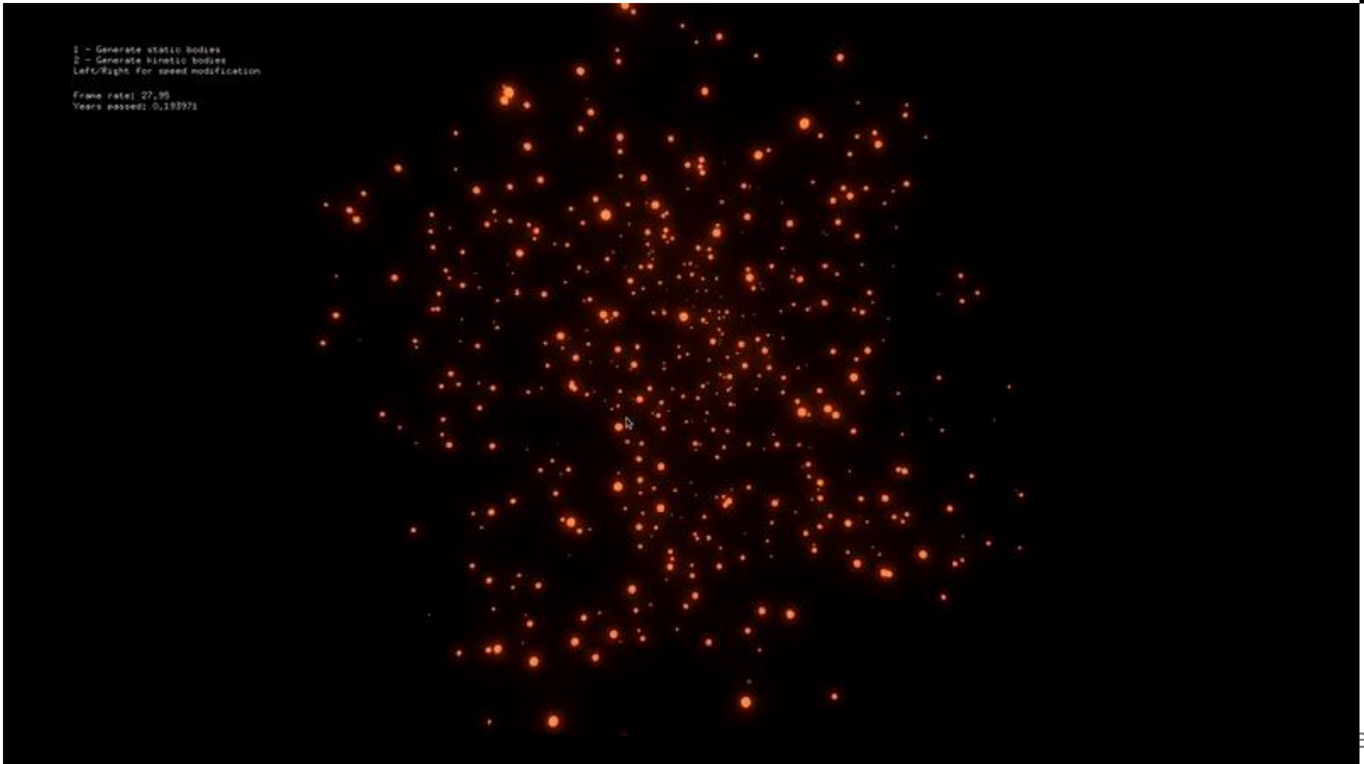
Simulation – Summary

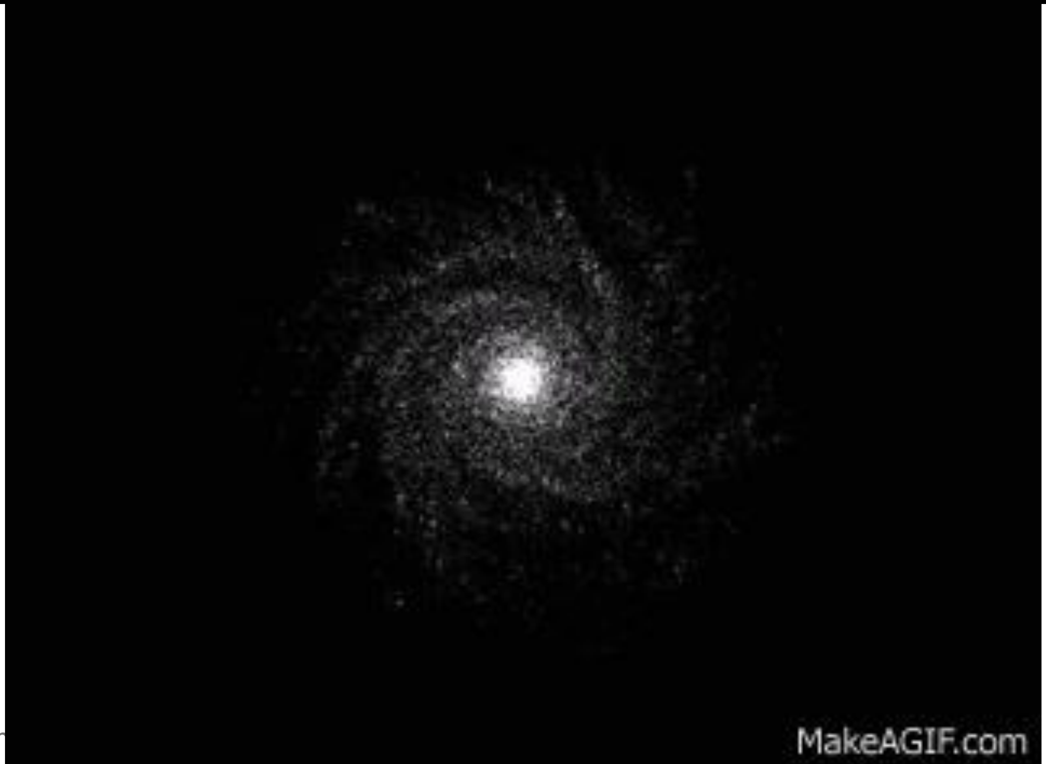
- If we want to simulate the projectile we need to
  - Set the initial position and velocity: $(x_x, x_y, x_z), (v_x, v_y, v_z)$
  - Sum all the forces applied to the particle $\sum F_i = F_g = mg$ (only gravity here)
  - Find the acceleration $a = \frac{F_g}{m} = g = 9{,}81\,\frac{m}{s^2}$ (using only the gravity force)
  - For each frame $t+1$:
    - Integrate a to find the velocity: $v_{t+1} = \int \frac{F_g}{m} dt = \int g dt = gt + v_t$
    - Integrate v to find the position: $x_{t+1} = \frac{1}{2} g t^2 + v_t t + x_t$
    - Update the position and velocity of the projectile
    - Draw the new particle in OpenGL

N-Body Simulation



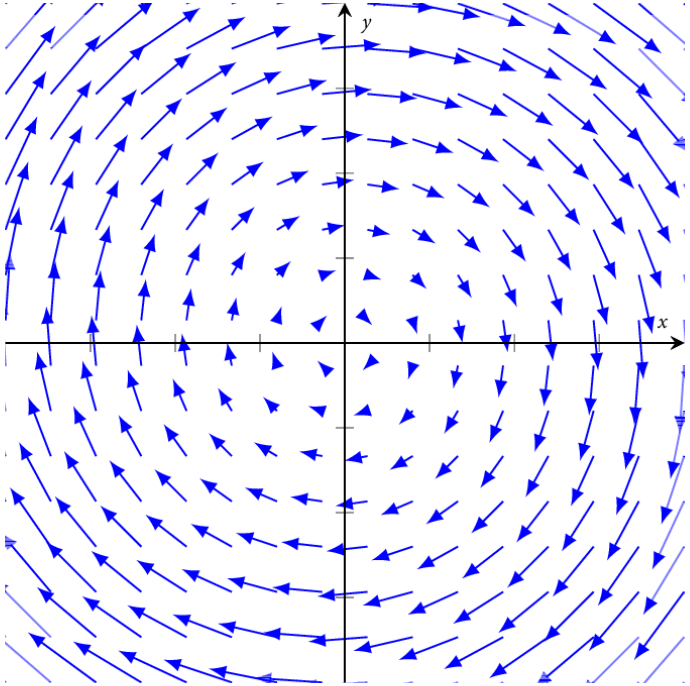


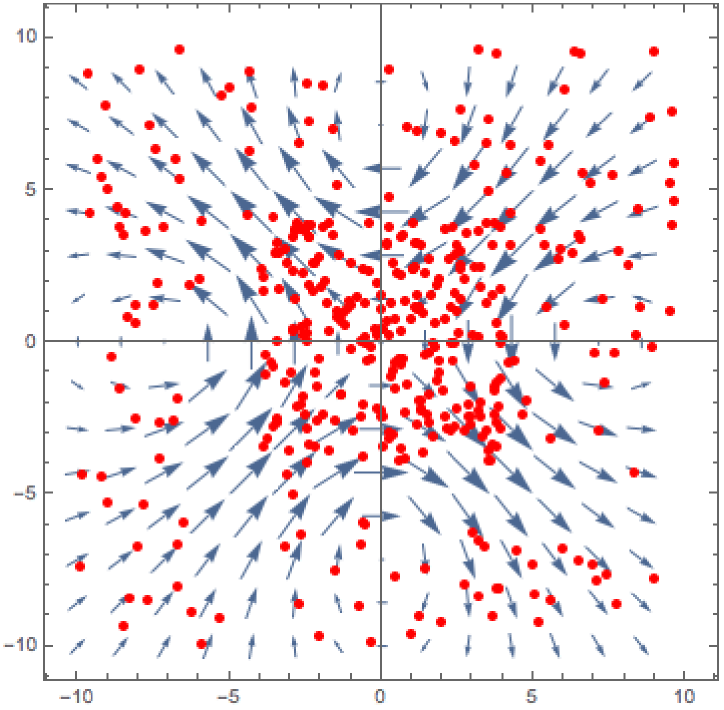


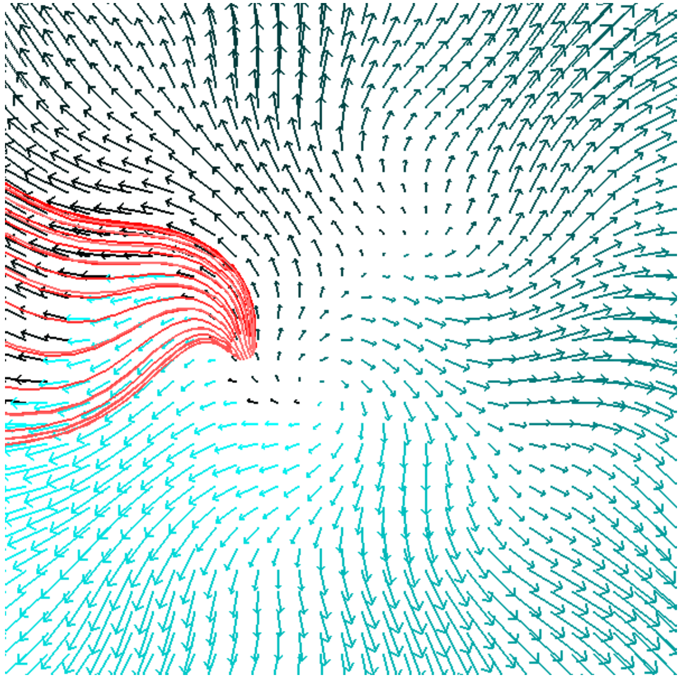


Vector Fields

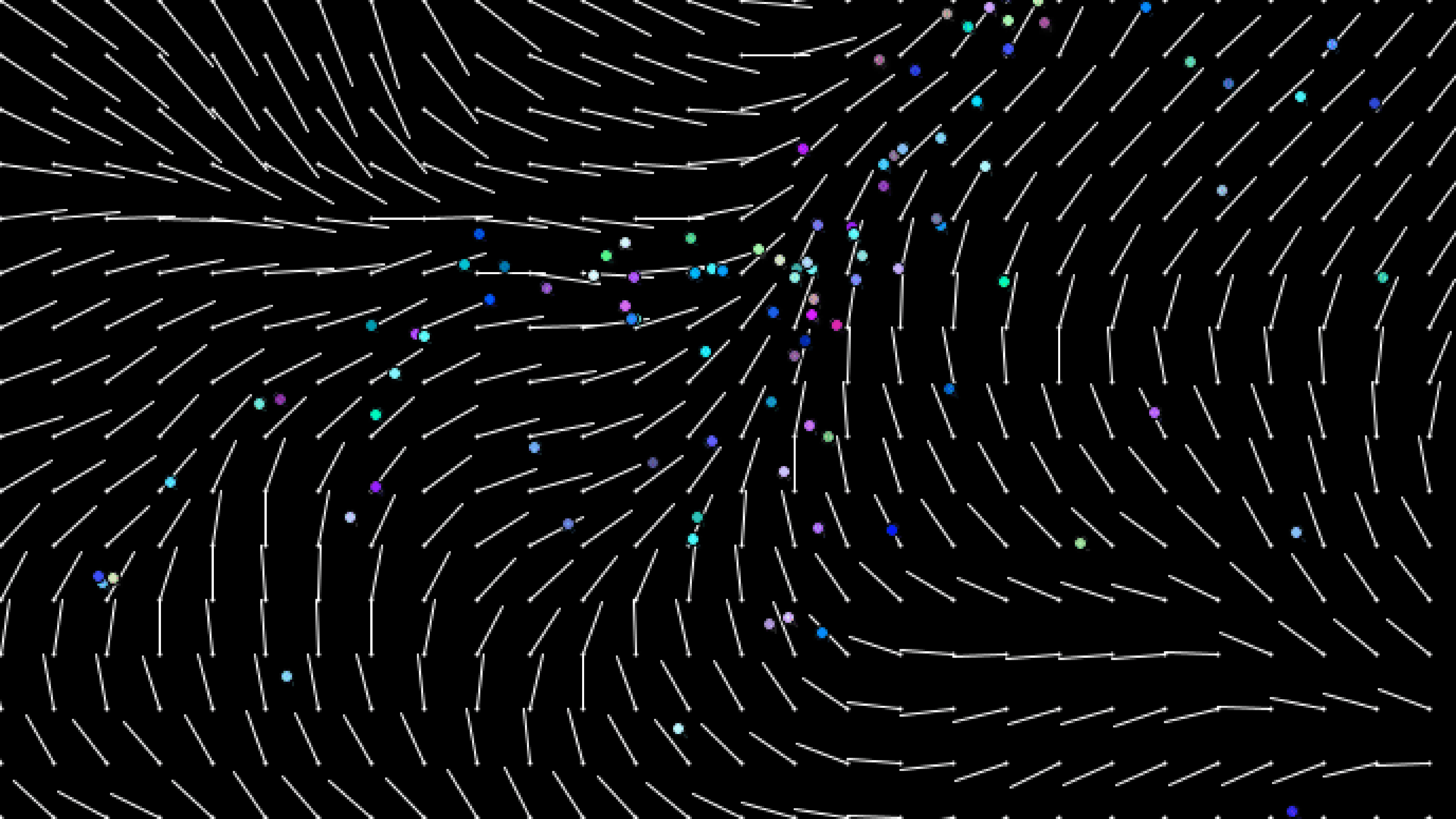
- Particle motion can also be given/affected by a function that takes a 3D position and returns a 3D velocity, i.e., a vector field.





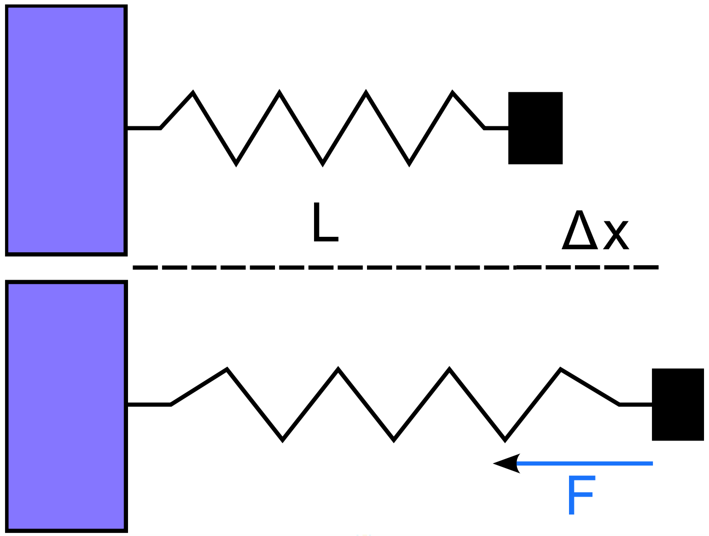


- In this case, each point in space gives a force, we only need to add it to the total forces on the particle

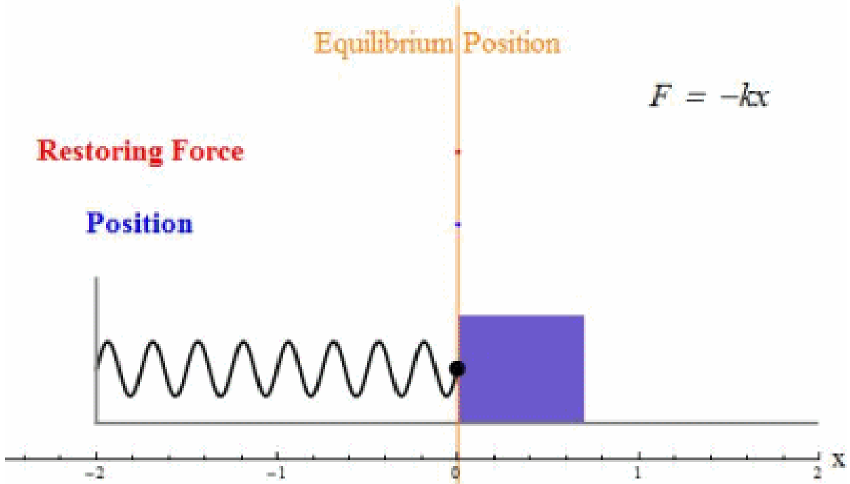


1D Springs With Fixed Object

- Springs are powerful to model complex structures (eg: Hairs, clothes, jello, ..)
- They connect two particles together, creating a mass-spring system.
- Springs have a rest length L and a stiffness constant k

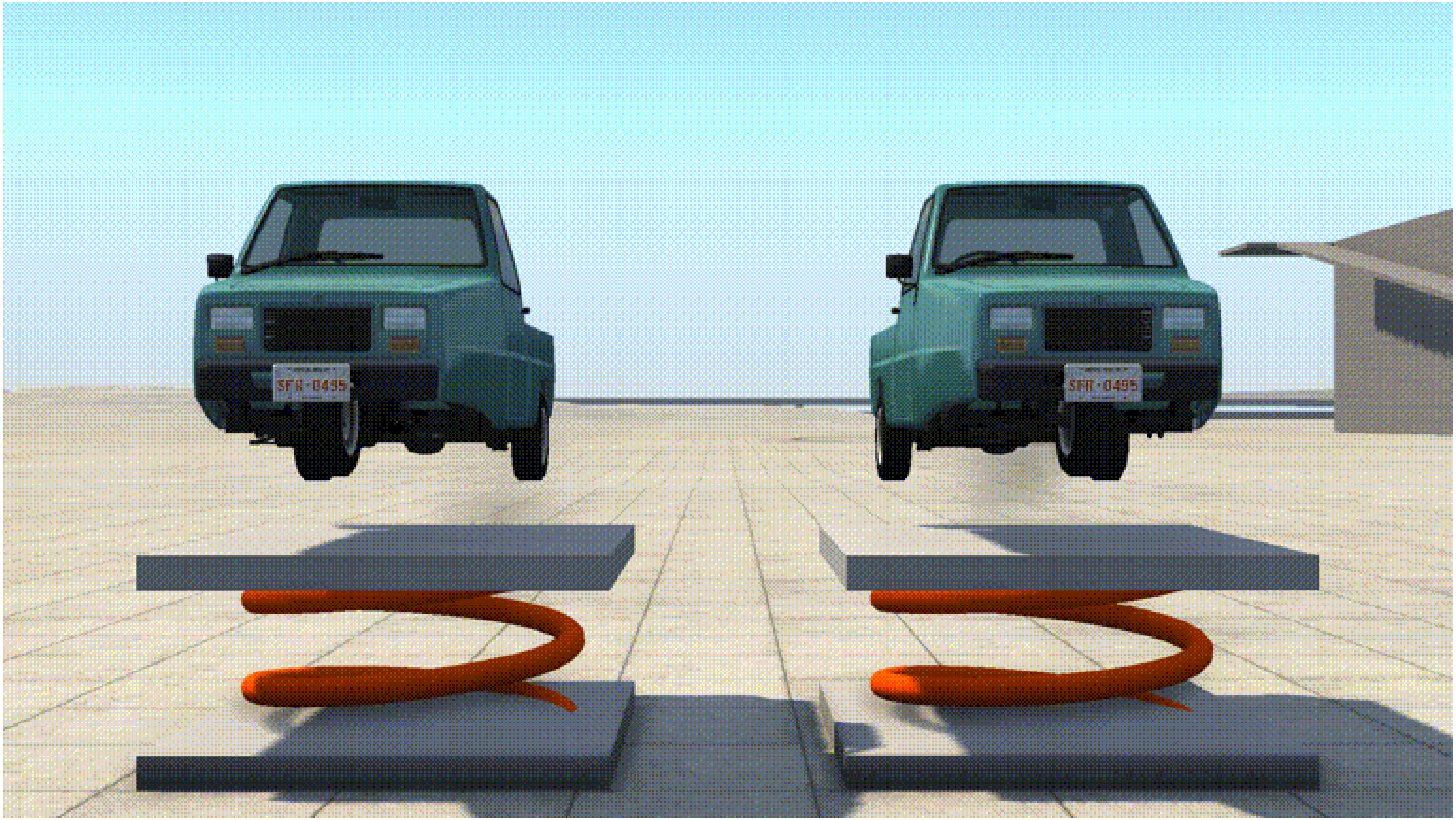


- Hooke's law:
  - The restoring force is linearly proportional to the amount of displacement from the rest length. It acts in opposite direction to the displacement.
  - $F_s = -k\Delta x$



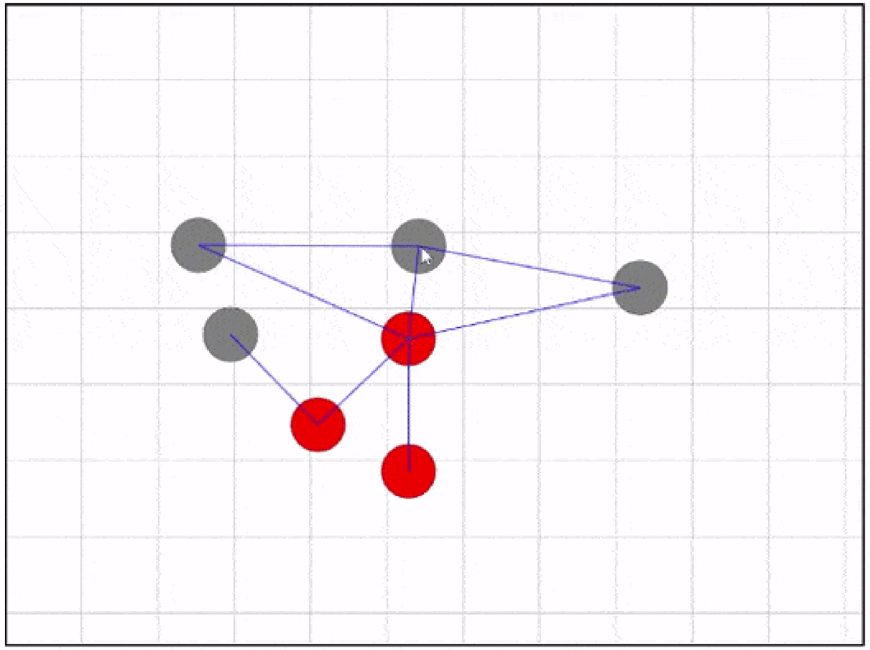
1D Springs With Fixed Object

What's different?



Spring Between Two Particles

- Two mobile particles move relative to each other.
- They exert to each other opposite forces !

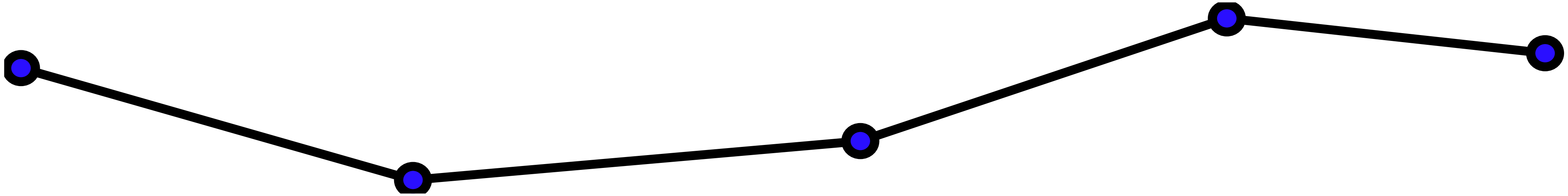


  - $F_1 = -F_2 = -k\left(\left|\left|X_1 - X_2\right|\right| - L\right)\frac{X_1 - X_2}{\left|\left|X_1 - X_2\right|\right|}$

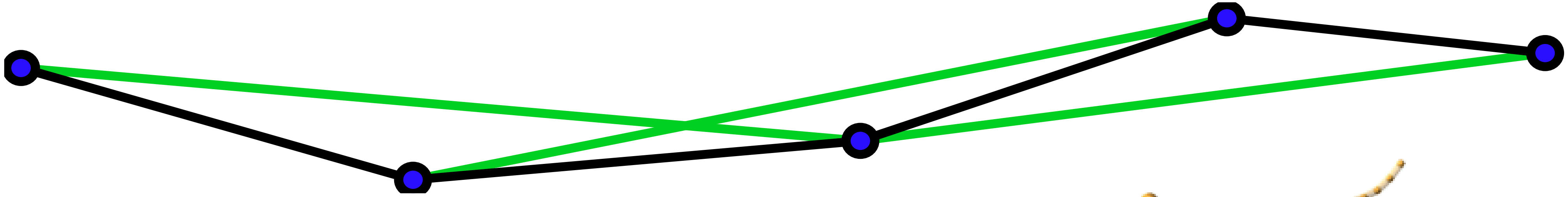
- When we add a spring force, the integral to find the velocity and position seems more complex, but when we evaluate our force expression by plugging in position values, it just becomes a scalar, not very complex to integrate.
- However, stiff objects require a higher discretisation of time for stability, or we would observe very high forces applied that decrease not fast enough.

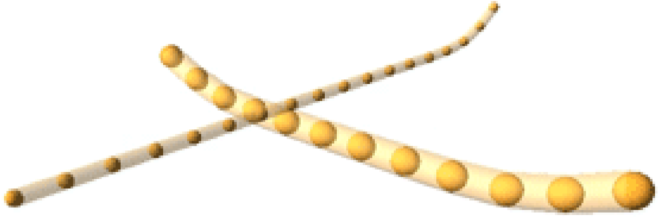
Application Of Springs – Hairs

- To model a single hair, we can chain together several particles with springs

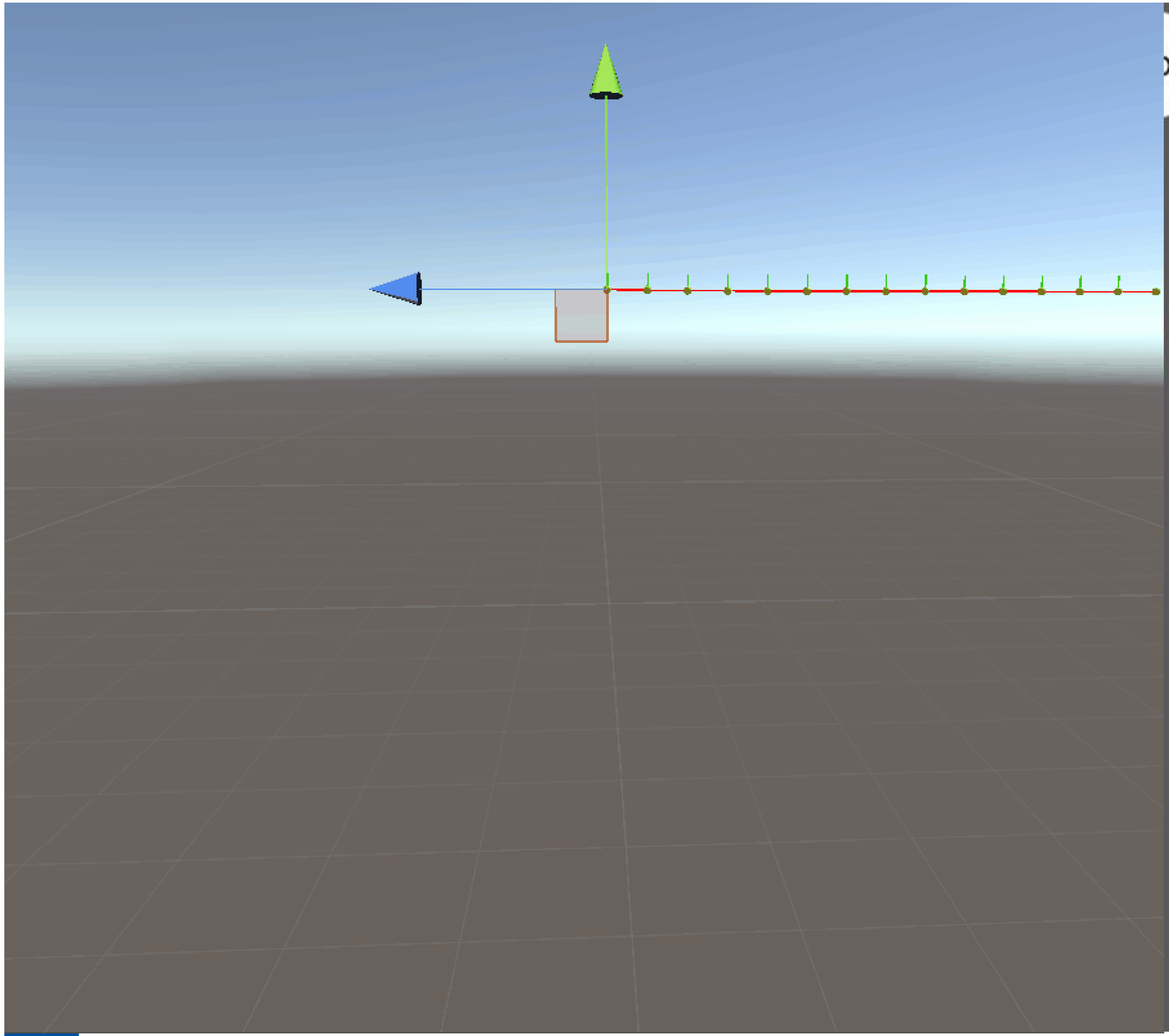


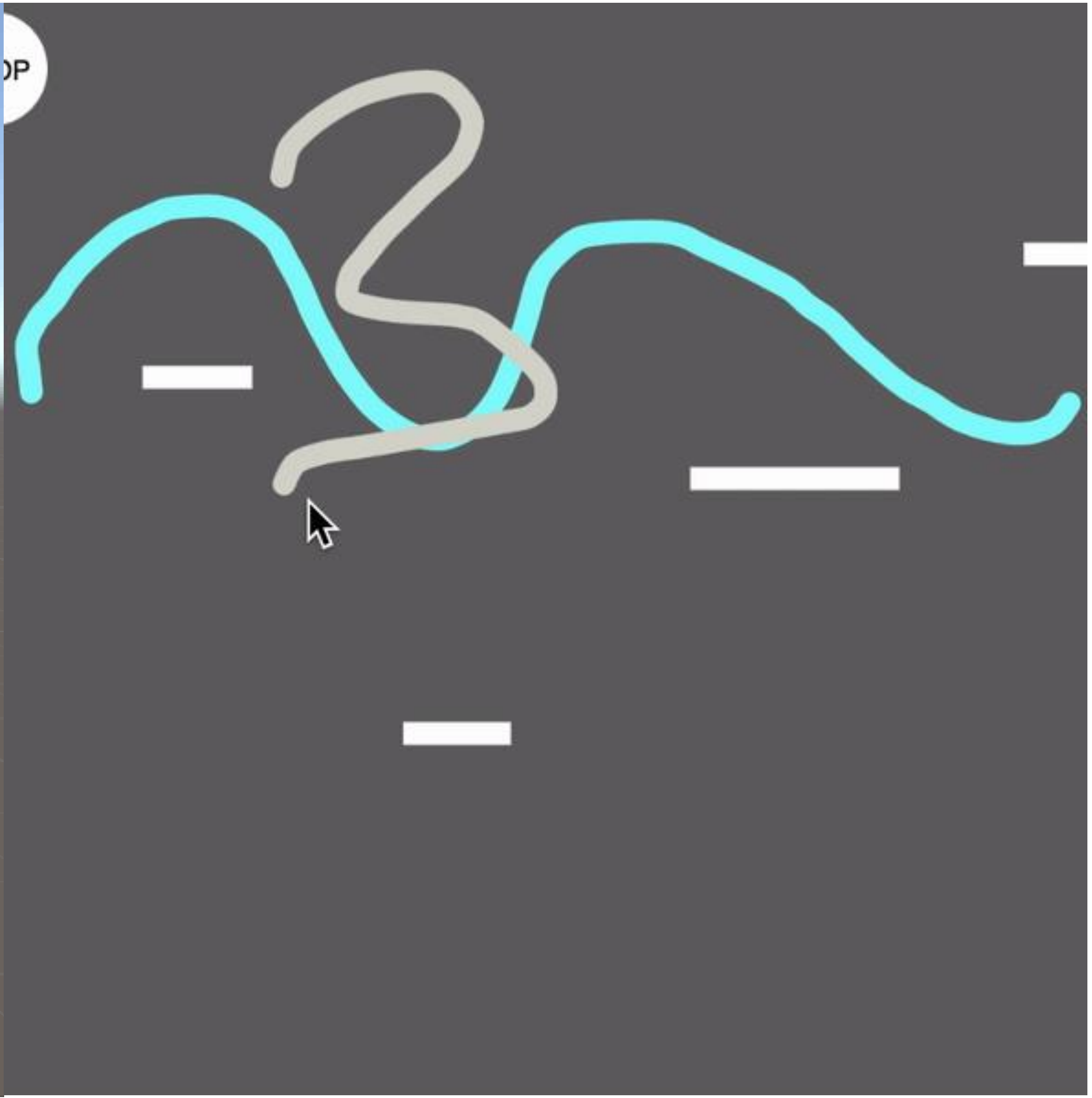
- However, doing this could collapse the hair
  - We improve our model by adding alternating springs among particles, discouraging the hair from collapsing and allowing better bending.

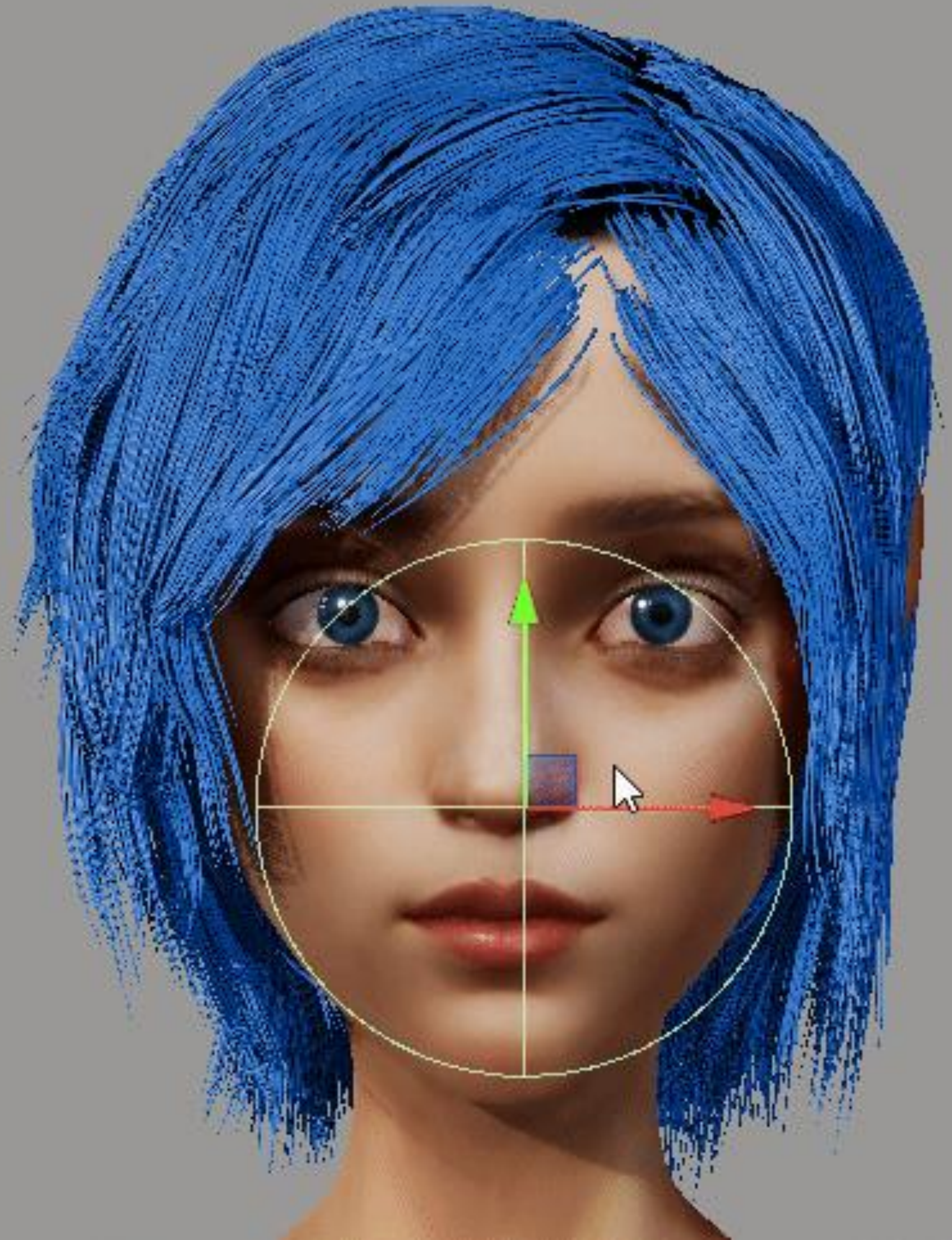




1D springs

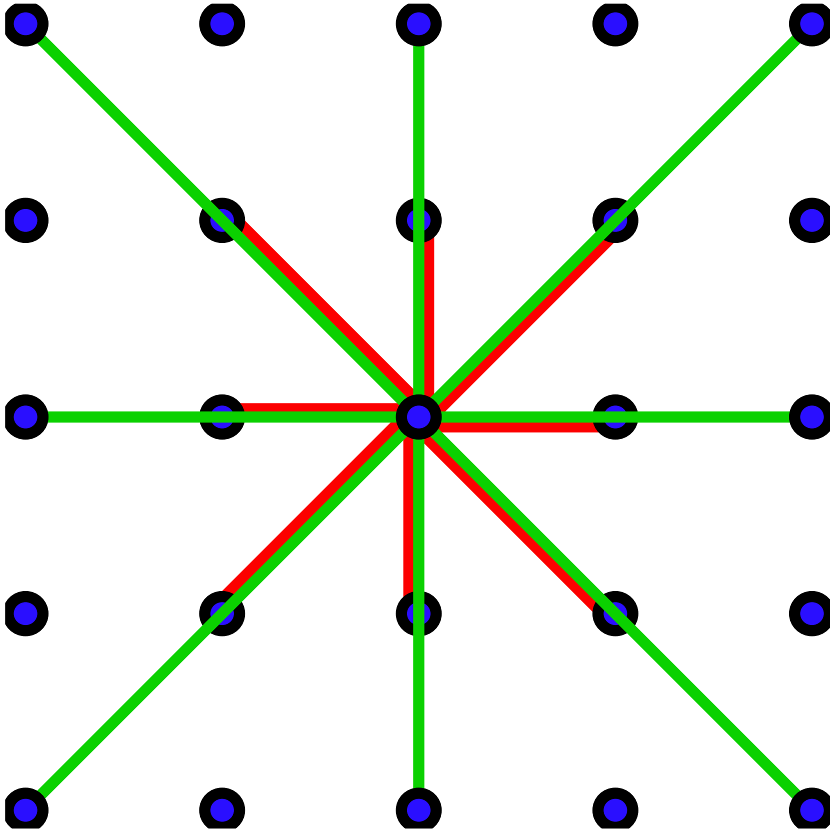


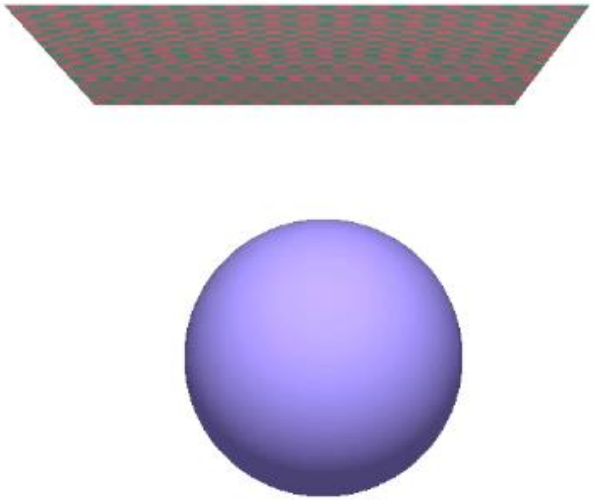




Application Of Springs – Cloth

- We can use a regular grid of particles to model a cloth.
  - Red springs model stretch/shearing
  - Green springs model bending

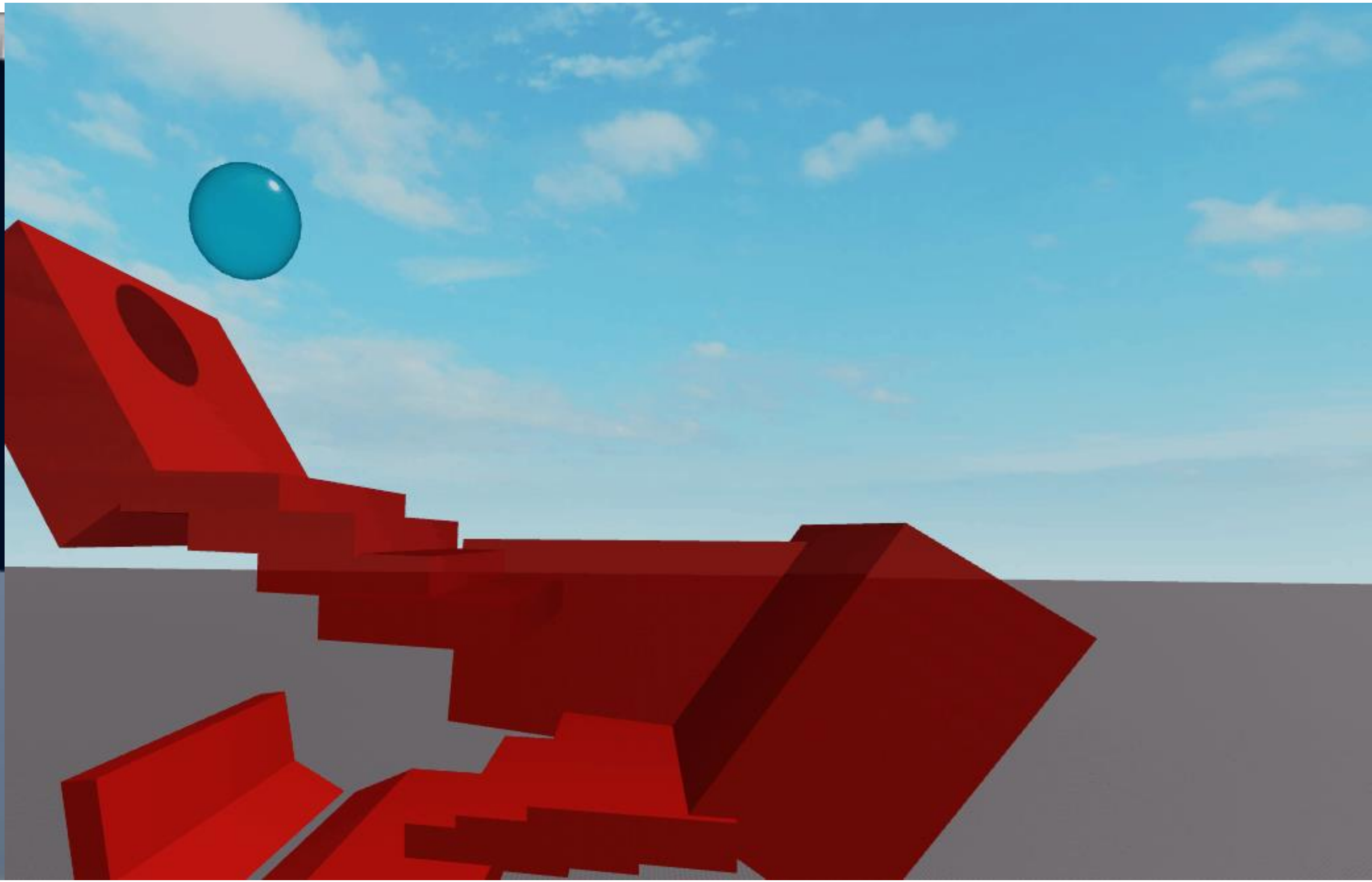




Springs for Soft Body Dynamics

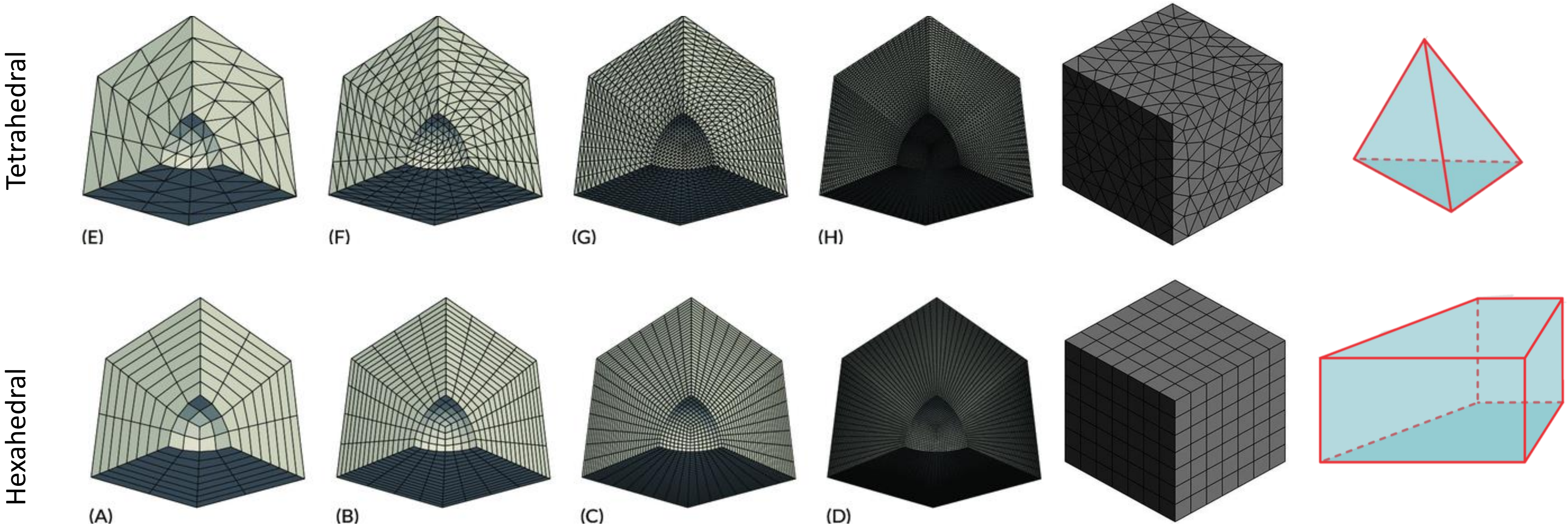
- A 3D network of springs forms a model for volumetric deformable objects

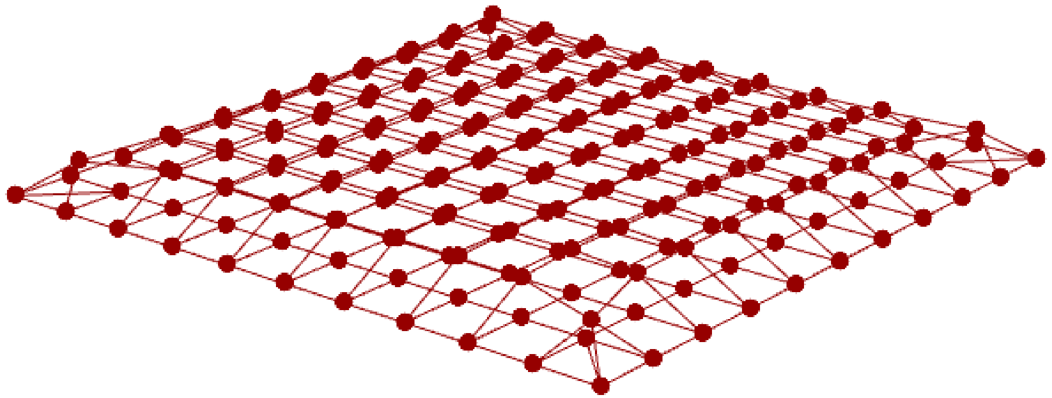
- Allows to use the mesh vertices as particles, or to have jelly objects !



Springs for Soft Body Dynamics

- Typically, we connect them using a tetrahedral or hexahedral mesh





Problems With Springs

- In cloth simulations, we added two kind of springs, one for stretch and the other for bending.
- However, they interact in non trivial ways, it is not easy to have a good model for cloth.

- If we have a finer grid resolution (particle density), the behavior is not consistent.
- A mass-spring system does not converge

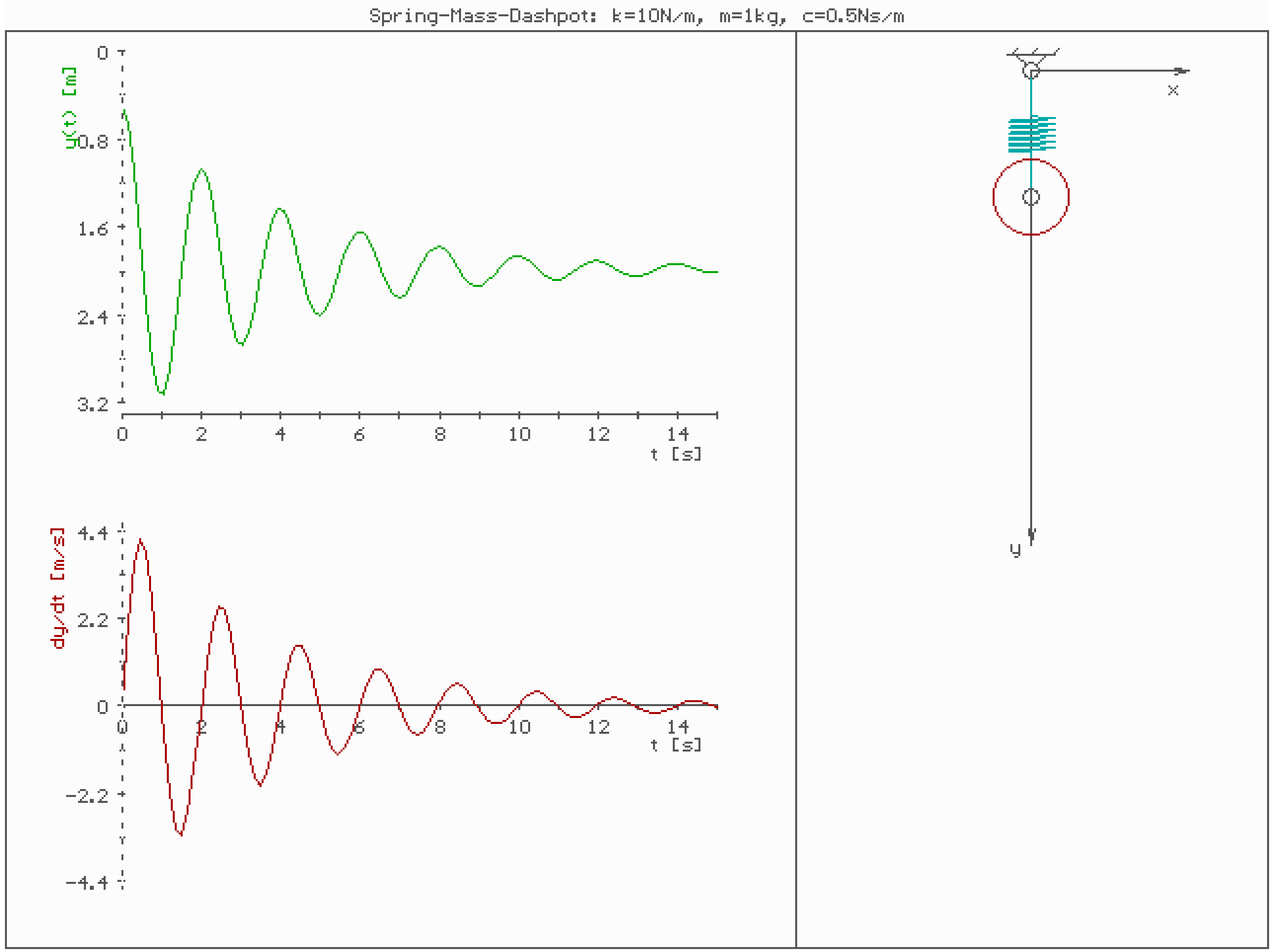
- Other approaches exist, such as finite difference, finite elements, finite volumes, you may have encourtered them in other lectures

Damping / Viscosity

- Approximates internal friction or energy losses.
- Gradually slows the motion, avoids oscillations forever

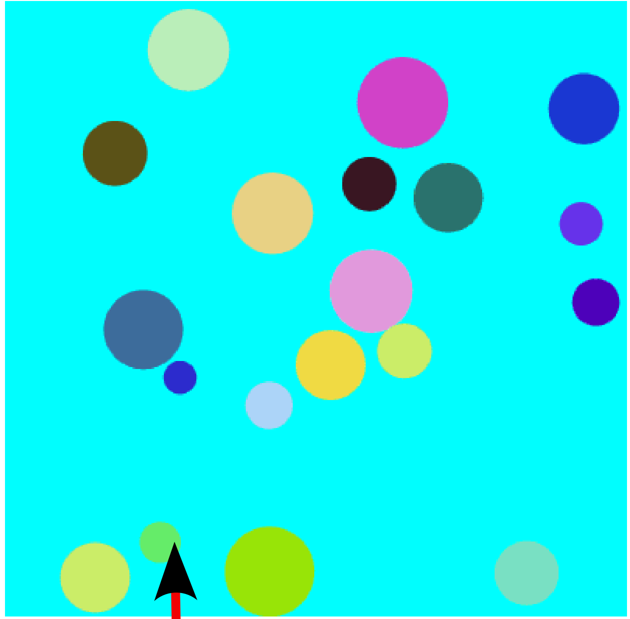
- Damping usually:
  - opposes the motion of the system and
  - is proportional to velocity
- Eg: $F_d = -k_d v(t)$
  - It is very similar to a spring

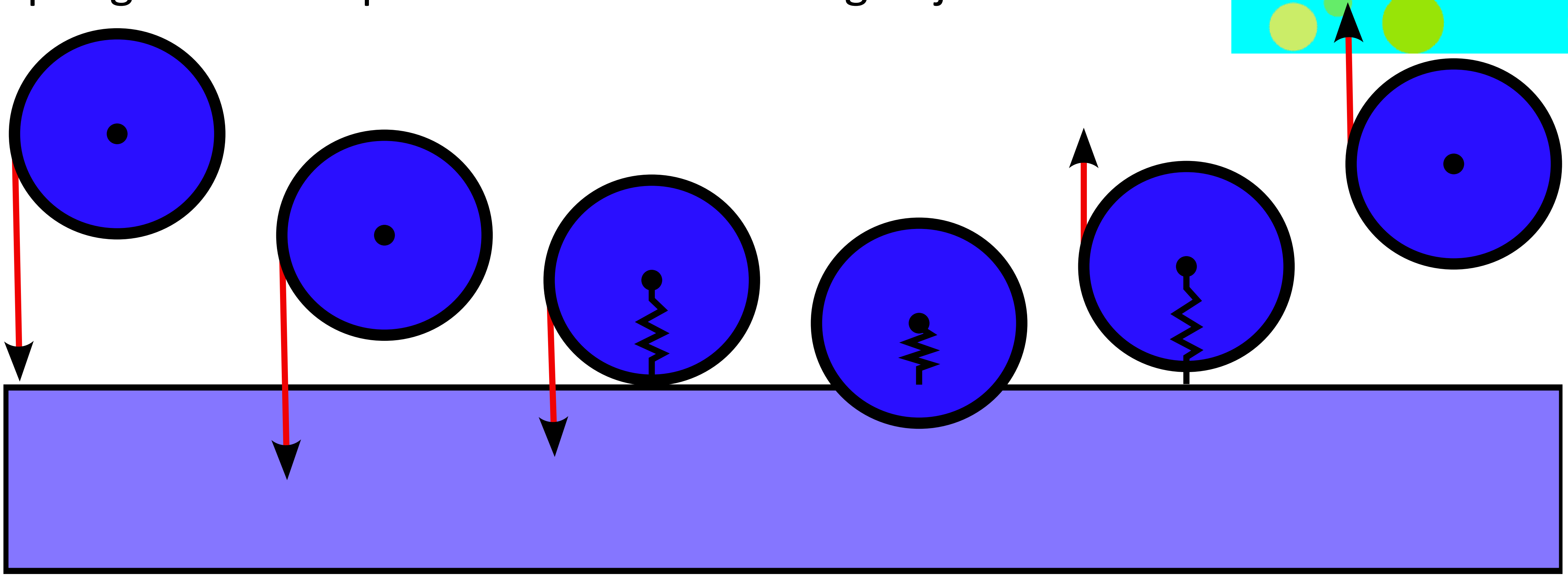
- Other models for damping and viscosity exists depending on the application physics.



Collisions

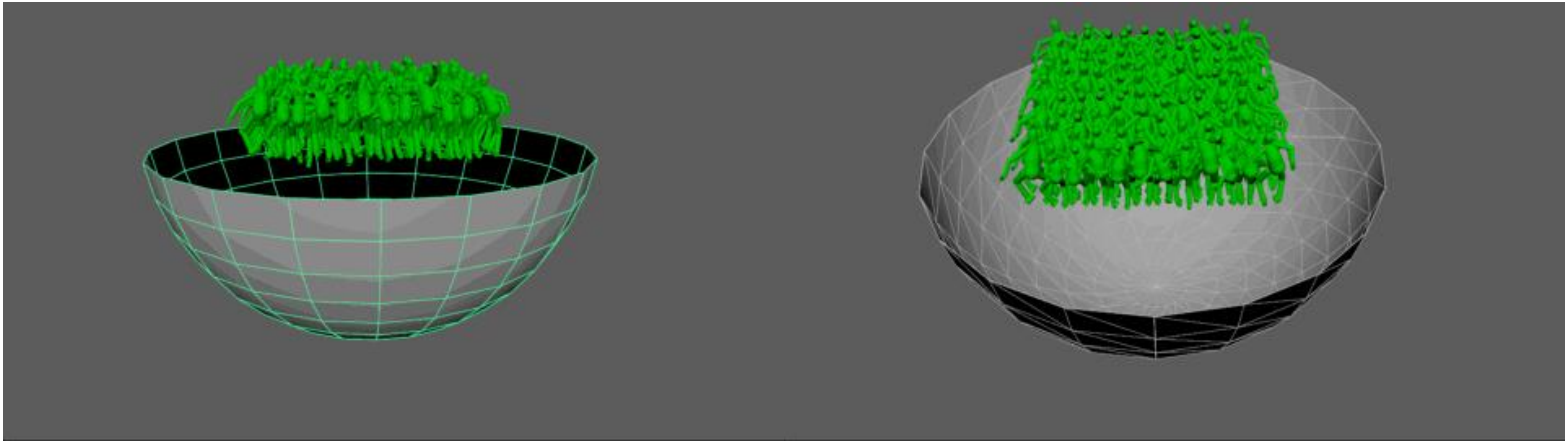
- Objects can collide
- we need to push them back in different directions.
- A common way to do this is to set temporary repulsive springs between particles of two colliding objects





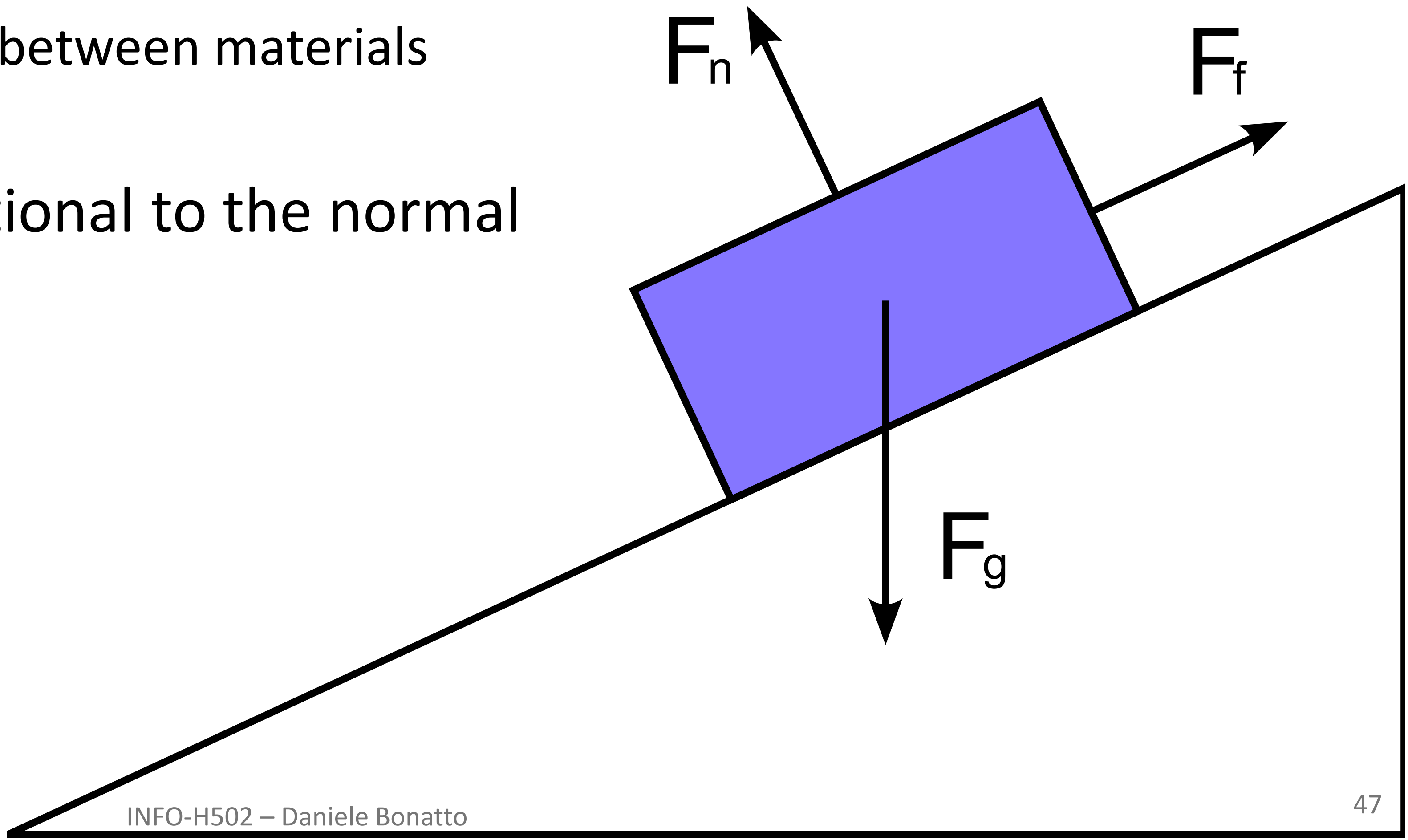
Collisions

- Sometimes, the discretization step in the integral is not fine enough !
- Or the mesh has not enough vertices !



Contact And Friction

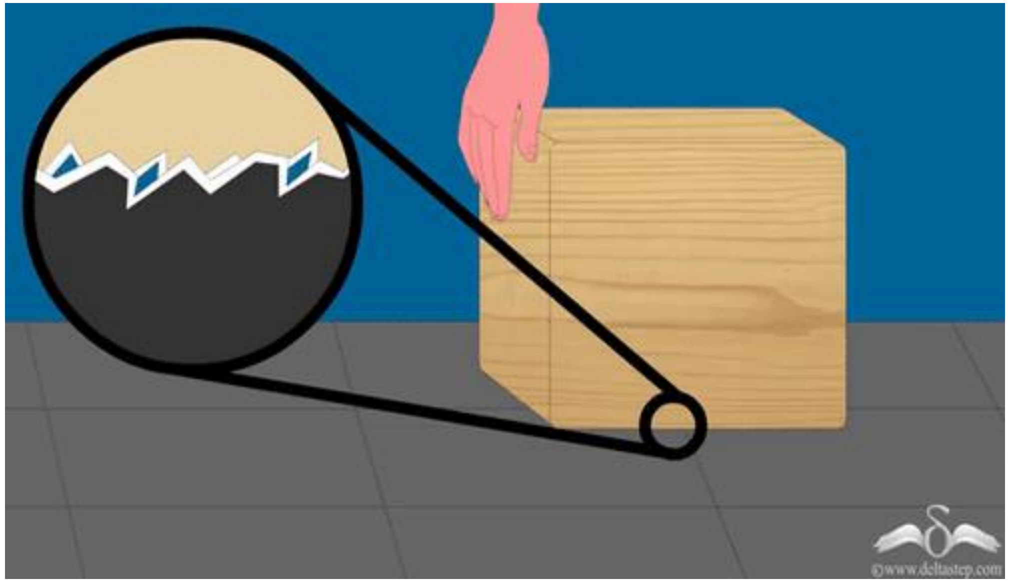
- When a body is over an inclined surface, it will have three forces:
  - F_g: The force of gravity
  - F_n: The normal force, opposing gravity
  - F_f: The friction force,
    that resist relative sliding between materials



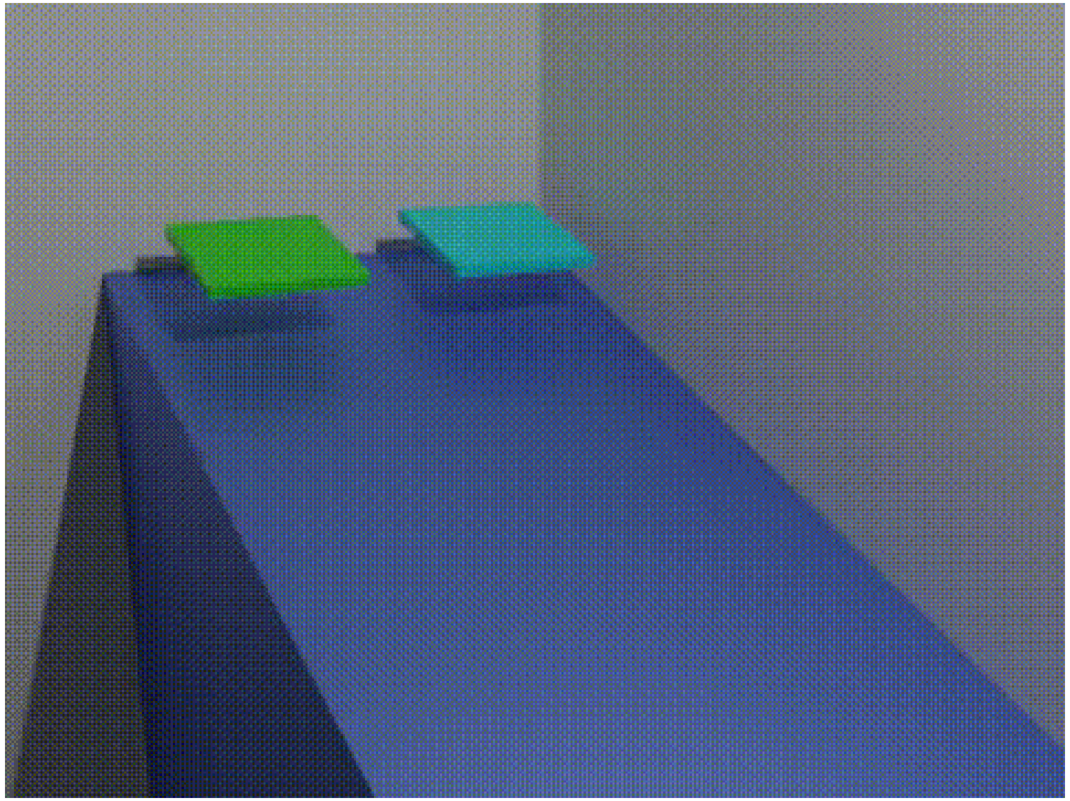
- The friction force is proportional to the normal
  - $F_f = \mu F_n$

Contact And Friction

- The friction force is generated by small disparities between materials

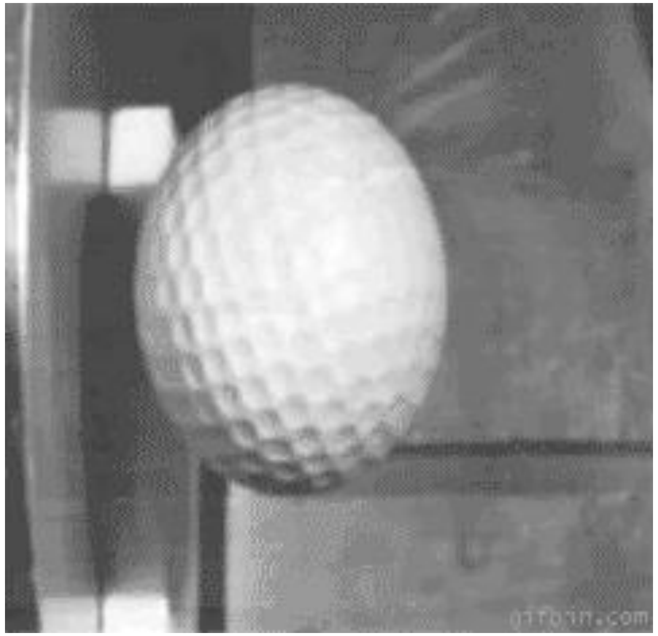


- Depending on the coefficient, we have more or less friction



Rigid Bodies

- Simulating very stiff objects with particles is problematic
  - Explicit methods: need very small steps to satisfy stability
  - Implicit methods: equations become ill-conditioned
  - Deformations happen very fast, but are not **necessarily** visually interesting for the application



Golf ball at 150k fps

- For stiff objects, we are only interested in their **position** and **rotation** !
- The position follows approximately the same treatment that for a particle, but we need more theory for rotations.

Rigid Bodies - Position

- We consider a rigid body made of particles, each with a mass m_i
- This body has a total mass $M = \sum m_k$
- And a center of mass $C = \frac{1}{M}\sum m_k r_k$

- Translating a rigid body consists in only moving its center of mass, all the other points in the body are fixed relatively to that point, and are moved in the same way.
- It follows a motion: $C = T + RC_0$

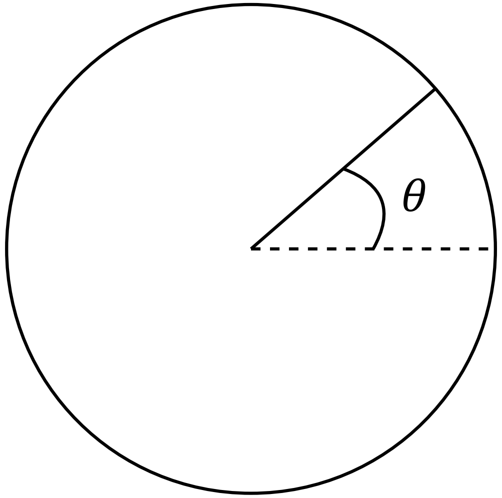
- For translations, we solve the same equations as for a particle, but this time applied to the center of mass.

Comparison Linear and Rotational Motion

Name	Linear	Rotational	Name
Mass	m	$M = \sum m_k, C = \frac{1}{M} \sum m_k r_k$	Center of mass
Conservation of linear momentum if no external forces act		Conservation of angular momentum if no external torques act	

Rigid Body – Rotational movements

- With linear motion we had x as the displacement
- In rotational motion we have θ as the angle of displacement



Comparison Linear and Rotational Motion

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Angular Velocity

- With linear motion, we had

$$v = \dot{x}$$

- However, in rotations, we cannot simply derive the angle of rotation θ
- This gives us only the angular velocity

$$\omega = \dot{\theta}$$

Comparison Linear and Rotational Motion

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Angular Velocity

- With linear motion, we had

$$v = \dot{x}$$

- However, in rotations, we cannot simply derive the angle of rotation θ
- This gives us only the angular velocity

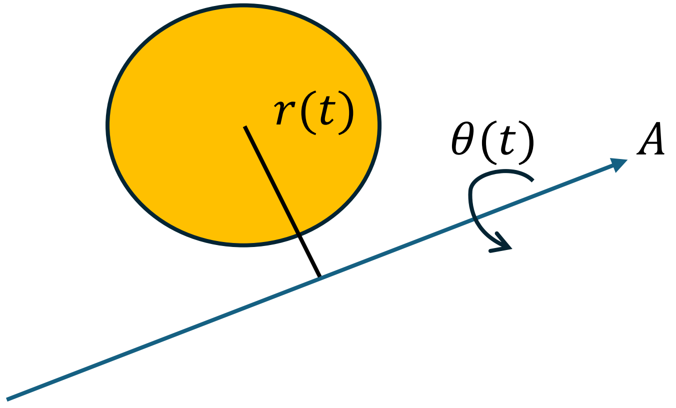
$$\omega = \dot{\theta}$$

- But the body rotates around some axis.
- Meaning that all the points on the rigid body do not move at the same angular velocity !

- How to consider that ?

Rigid Body – Rotational Movements

- What do we have ?
  - A rigid body with a center of mass C
  - An off-centered rotation axis A
  - The angle of rotation θ
  - The distance r between the center of mass and the rotation axis A



- The speed at which the points are rotating is proportional to the distance r from the axis of rotation
  - $v(t) = |\omega . r|$

Try it on a rotating chair by extending your arms
Why does it work ?

- But this does not tell the direction of rotation
- We can find it using a vector product!
  - $\dot{r} = v = \omega \times r$

Comparison Linear and Rotational Motion

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		$v = \dot{r} = \omega \times r$	Linear velocity (off axis)
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Acceleration

- The same apply for the acceleration
- In linear motion, we had

$$a = \dot{v}$$

- With our rotation, we obtain the angular acceleration

$$\alpha = \dot{\omega}$$

Comparison Linear and Rotational Motion

Name	Linear	Rotational	Name
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Acceleration	$a = \ddot{x} = \dot{v}$	$\alpha = \dot{\omega}$	Angular Acceleration
Conservation of linear momentum if no external forces act		Conservation of angular momentum if no external torques act	

Acceleration

- The same apply for the acceleration
- In linear motion, we had

$$a = \dot{v}$$

- With our rotation, we obtain the angular acceleration

$$\alpha = \dot{\omega}$$

- But, in a rotating rigid body, the acceleration must depend on the distance from the axis of rotation

Acceleration

- We have the off-axis velocity $\dot{r} = \; v(t) = \omega \times r$
- let's find the acceleration

$$\alpha = \dot{v} = \dot{\omega} \times r + \omega \times \dot{r}$$
$$\alpha = \dot{v} = \dot{\omega} \times r + \omega \times (\omega \times r)$$

Comparison Linear and Rotational Motion

Name	Linear	Rotational	Name
Mass	m	$M = \sum m_k, C = \frac{1}{M} \sum m_k r_k$	Center of mass
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Acceleration	$a = \ddot{x} = \dot{v}$	$\alpha = \dot{\omega}$	Angular Acceleration
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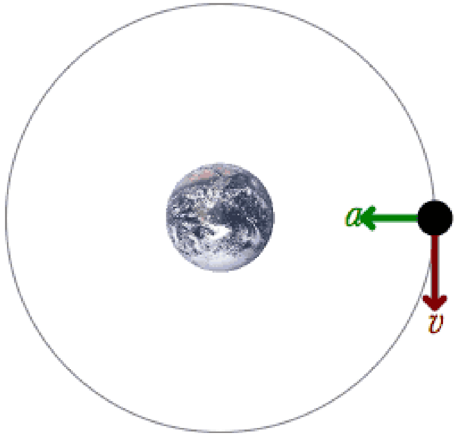
Conservation of linear momentum if no external forces act		Conservation of angular momentum if no external torques act	

Acceleration – Centrifugal force

$$\alpha = \dot{v} = \dot{\omega} \times r + \omega \times (\omega \times r)$$

- Let's analyze an example with constant velocity ω



- As the velocity ω is constant, $\dot{\omega} \times r = 0$
- The term $\omega \times (\omega \times r)$ is not zero, and represents the *tension* of the string between the axis of rotation and the particle.
- The particle experiences an opposite force:
- $F_{centrifugal} = m\alpha = -m\big(\omega \times (\omega \times r)\big)$
- It causes the object to move away from the axis of rotation

Angular moment – Torque

- With linear motion we had the linear momentum
  - $p = mv$
- It expresses the difficulty to put into movement a body of mass m

- Its rotational analog is called angular momentum
  - $L = r \times p$
- This time, the difficulty depends on the rotation axis r

Comparison Linear and Rotational Motion

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Displacement	x	θ	Angle
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Linear Momentum	$p = mv$	$L = r \times p$	Angular Momentum
Conservation of linear momentum if no external forces act		Conservation of angular momentum if no external torques act	

ULB

Force

- In linear movement, we could differentiate the linear momentum to get the force

$$F = \dot{p} = m\dot{v} = ma$$

- Again, this depends on the axis of rotation of the body for rotational movement !

Angular moment – Torque

$$\frac{d}{dt}\begin{pmatrix} r(t) \\ ? \end{pmatrix} = \begin{pmatrix} \omega(t) \times r(t) \\ ? \end{pmatrix}$$

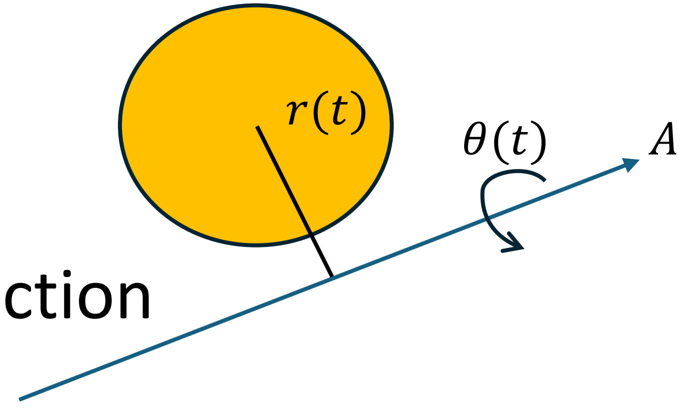
$$L = r \times p$$

- Differentiating both sides:

$$\dot{L} = \dot{r} \times p + r \times \dot{p}$$

- Since $\dot{r} = v$, $\dot{r}$ and $p = mv$ points in the same direction

$$\dot{L} = r \times \dot{p} = r \times m\dot{v}$$



- The vector $m\dot{v}$ is equal to the total force on the particle ! ($F = ma$)

$$\dot{L} = r \times F$$

- This quantity is defined as the torque (analog to force for rotations)

$$\tau = \dot{L} = r \times F$$

Comparison Linear and Rotational Motion

Name	Linear	Rotational	Name
Mass	m	$M = \sum m_k, C = \frac{1}{M}\sum m_k r_k$	Center of mass
Displacement	x	θ	Angle
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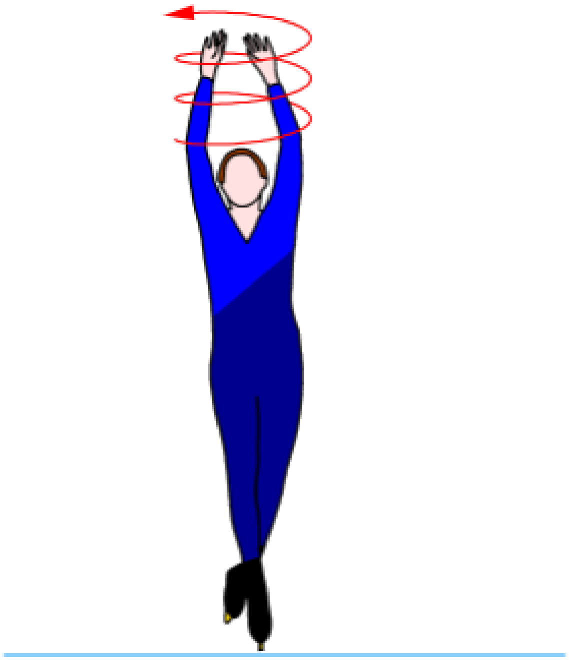
Conservation of linear momentum if no external forces act		Conservation of angular momentum if no external torques act	

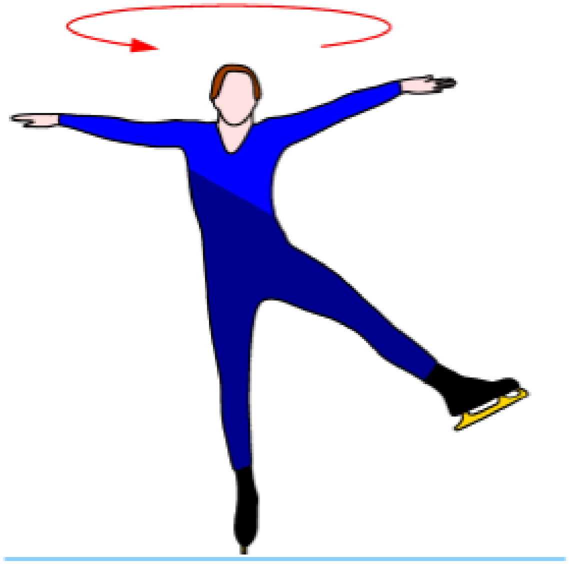
ULB

Resisting Movement – Linear and Rotational

- For a linear motion, the mass m determines how hard it is to accelerate an object $(F = ma)$.

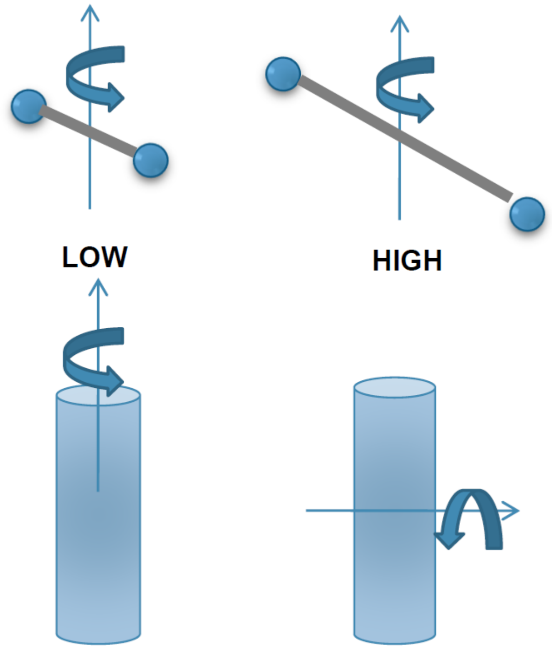
- How do objects resist rotational acceleration ?
- Try:





Resisting Movement – Linear and Rotational

- For a linear motion, the mass m determines how hard it is to accelerate an object ($F = ma$).



- How do objects resist rotational acceleration ?
- The resistance depends on two things:
  - The **mass** of the object
  - How the **mass is distributed** relative to the axis of rotation
- For rotational motion, we define an equivalent quantity to m that accounts for the rotational axis and the mass distribution.
- This quantity is the **moment of inertia**.

Ice Skating Example

- The **moment of inertial** is a scalar that quantifies how mass is distributed relative to a specific axis of rotation. For a discrete system of point masses:

$$I = \sum_i m_i r_i^2 \in \mathbb{R}$$

- Where:
  - m_i is the mass of the i-th point.
  - r_i is the perpendicular distance of the i-th point from the axis of rotation
- For a continuous mass distribution, we use an integral:

$$I = \int_V \rho(r) r^2 dV = \int_V r^2 dm \in \mathbb{R}$$

- Where:
  - $\rho(r)$ is the mass density at position r
  - r the perpendicular distance from the axis of rotation.

e.g., $I_z = \int_V (x^2 + y^2) dm$

Comparison Linear and Rotational Motion

Name	Linear	Rotational	Name
Mass	m	$M = \sum m_k, C = \frac{1}{M}\sum m_k r_k, I$	Center of mass, Moment of inertia
Displacement	x	θ	Angle
Velocity	$v = \dot{x}$	$\omega = \dot{\theta}$	Angular Velocity
		$v = \dot{r} = \omega \times r$	Linear velocity (off axis)
Acceleration	$a = \ddot{x} = \dot{v}$	$\alpha = \dot{\omega}$	Angular Acceleration
		$\alpha = \dot{\omega} \times r + \omega \times (\omega \times r)$	Acceleration (off axis)
Linear Momentum	$p = mv$	$L = \mathrm{r} \times p$	Angular Momentum
Force	$F = ma$	$\tau = \mathrm{r} \times F$	Torque
Conservation of linear momentum if no external forces act		Conservation of angular momentum if no external torques act	

- The moment of inertia determines how much torque τ is required to achieve an angular acceleration α:

$$\tau = I\alpha$$

- We don’t give the demonstration here, it is mostly a manipulation of the equation
- But, we see that it makes intuitive sense $F = ma$
  - The torque is the equivalent to the force
  - The moment of inertia the equivalent to the mass
  - The angular acceleration the equivalent to the acceleration

Comparison Linear and Rotational Motion

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Force	$F = ma$	$\tau = \mathrm{r} \times F = I\alpha$	Torque
Conservation of linear momentum if no external forces act		Conservation of angular momentum if no external torques act	

- The moment of inertia is a **scalar** and works well when rotation is about a fixed axis aligned with the coordinates system.

- However, in more complex cases (e.g., rotation about arbitrary axes or in 3D), a scalar is insufficient. We need a framework that:
  - Accounts for the **direction** of rotation.
  - Handles cases where the **axis of rotation** is not aligned with principal axes.

- This leads to the **Inertia Tensor**, a more general representation.

- The inertia tensor I is defined as:

$$I_{jk} = \int_V \rho(r)\left(r^2\delta_{jk} - r_j r_k\right)dV$$

- Where
  - r_j, r_k are the components of $r(t)$ in the j-th and k-th directions
  - δ_{jk} is the Kronecker delta $\delta_{jk} = 1$ if $j = k$, otherwise $\delta_{jk} = 0$

- The proof is also not of the interest of those lectures and involves mostly massaging the $L = r \times p$ equation when we consider all the points in the rigid body.

Inertia Tensor

- Inertia Tensor is a property of an object that describes the mass distribution of a body
- Can be seen as a scaling factor for the effect of Torques on a body from different directions

$$I(t) = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} \in \mathbb{R}^{3\times 3}$$

- Diagonal terms are called moment of inertia $I_{xx} = M \int_V (y^2 + z^2)\, dm$
  - The scalar we previously described
- Other terms are products of inertia $I_{xy} = -M \int_V xy\, dm$
- Inertial tensor is invariant to tranlation, but changes with a body orientation

Inertia Tensor

- Inertia tensor changes depending of the orientation of the object
- But, we can compute it easily !

$$I(t) = R(t)I_{body}R^T(t)$$

- Where I_{body} is the neutral tensor of inertia of the body.

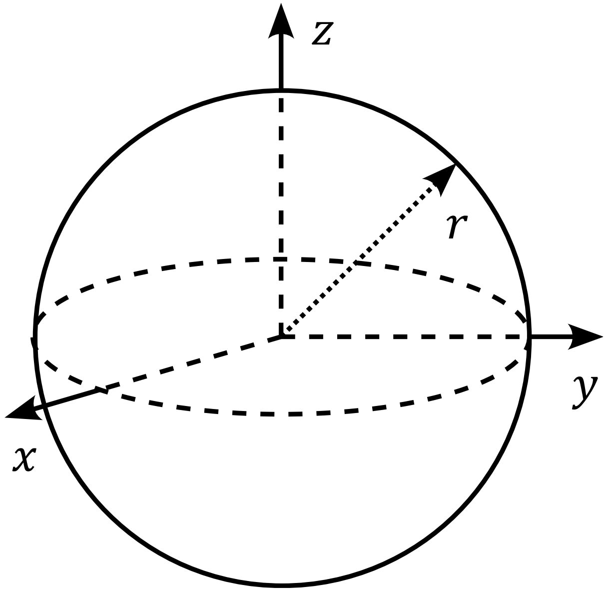
- To find I_{body}, we need to use the formula for the inertia tensor

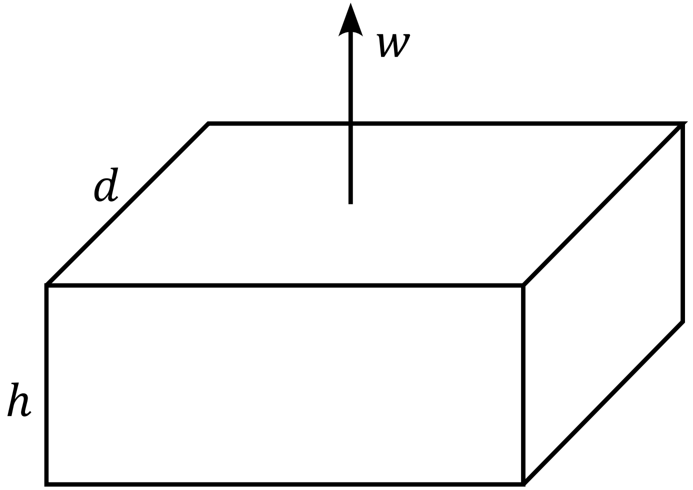
$$I_{jk} = \int_V \rho(r)\left(r^2\delta_{jk} - r_j r_k\right)dV$$

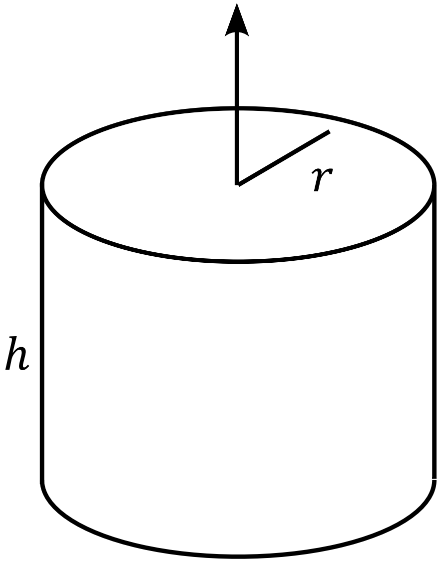
- And apply it to the particular geometry of our problem

Examples of Inertia Tensors

- Examples







$$I_{sph}(t) = \begin{pmatrix} \frac{2}{5}mr^2 & 0 & 0 \\ 0 & \frac{2}{5}mr^2 & 0 \\ 0 & 0 & \frac{2}{5}mr^2 \end{pmatrix} \quad I_{box}(t) = \begin{pmatrix} \frac{1}{12}m(h^2+d^4) & 0 & 0 \\ 0 & \frac{1}{12}m(u^2+d^2) & 0 \\ 0 & 0 & \frac{1}{12}m(w^2+h^2) \end{pmatrix} \quad I_{cyl}(t) = \begin{pmatrix} \frac{1}{12}m(3r^2+h^2) & 0 & 0 \\ 0 & \frac{1}{12}m(3r^2+h^2) & 0 \\ 0 & 0 & \frac{1}{12}mr^2 \end{pmatrix}$$

Inertia Tensor

- We still do not know how to convert between angular velocity ω and angular momentum $L(t)$

- A useful equation comes in hand:

$$L = I\omega$$

Its derivation is out of scope of those lectures, but its mostly vector products and simplifications

- It's the rotational counterpart of mass.
  - $L = I\omega$ vs $p = mv$

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Displacement	x	θ	Angle
Velocity	$v = \dot{x}$	$\omega = \dot{\theta}$	Angular Velocity
		$v = \dot{r} = \omega \times r$	Linear velocity (off axis)
Acceleration	$a = \ddot{x} = \dot{v}$	$\alpha = \dot{\omega}$	Angular Acceleration
		$\alpha = \dot{\omega} \times r + \omega \times (\omega \times r)$	Acceleration (off axis)
Linear Momentum	$p = mv$	$L = r \times p = I\omega$	Angular Momentum
Force	$F = ma$	$\tau = \mathrm{r} \times \mathrm{F} = I\alpha$	Torque
Conservation of linear momentum if no external forces act		Conservation of angular momentum if no external torques act	

Other Quantities

- We can define other useful Formulas:
  - Kinetic equations
  - Kinetic energy, and
  - Work
- They do not depend on the axis of rotation and can be defined directly as summed up in the following table.

Comparison Linear and Rotational Motion

Name	Linear	Rotational	Name
Mass	m	$M = \sum m_k, C = \frac{1}{M}\sum m_k r_k, I$	Center of mass, Moment of inertia
Displacement	x	θ	Angle
Velocity	$v = \dot{x}$	$\omega = \dot{\theta}$	Angular Velocity
		$v = \dot{r} = \omega \times r$	Linear velocity (off axis)
Acceleration	$a = \ddot{x} = \dot{v}$	$\alpha = \dot{\omega}$	Angular Acceleration
		$\alpha = \dot{\omega} \times r + \omega \times (\omega \times r)$	Acceleration (off axis)
Linear Momentum	$p = mv$	$L = r \times p = I\omega$	Angular Momentum
Force	$F = ma$	$\tau = \dot{L} = r \times F = I\alpha$	Torque
Kinematic equations	$v = v_0 + at$ $x = x_0 + v_0 t + \frac{1}{2}at^2$	$\omega = \omega_0 + \alpha t$ $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$	Kinematic equations
Kinetic Energy	$KE = \frac{1}{2}mv^2$	$KE = \frac{1}{2}I\omega^2$	Kinetic Energy
Work	$W = Fs$	$W = \tau\theta$	Work
Conservation of linear momentum if no external forces act		Conservation of angular momentum if no external torques act	

- You now have all the formulas to make rigid body motions !

- But.. What about fluids ?

Fluids

- Fluids have no prefered shape
- Always flow when force is applied
- Deform to fit their container
- Internal forces depend on velocities, not displacement/déformations

- Very short introduction to fluid animation
- It would need a complete course to cover it correctly INFO-H50X ?

Incompressible Navier-Stokes Equation

- Most fluids follow the incompressible Navier-Stokes equation
- Which is a set of partial differential equations that hold throughout the fluid.

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

- Let’s break it down !

Symbols

- First of all, the symbols !
- $\vec{u} = (u, v, w)$: Velocity of particles (traditionally u in fluid mechanics)
- ρ: Density of the fluid ($\approx 1000 kg/m^3$ for water)
- p: Pressure
- $\vec{g}$: Gravity ($9{,}81\ m/s^2$)
- ν: kinematic viscosity (How much the fluid resists deforming while it flows)

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$

$$\nabla \cdot \vec{u} = 0$$

- This equation is in reality 3 equations
- But what does it means ?

Incompressible Navier-Stokes Equation

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$

$$\nabla \cdot \vec{u} = 0$$

- The equation is actually the Newton second law $F = ma$ in disguise !
- It tells us how the fluid accelerates due to forces acting on it.

- Now, we break it down to its components to understand it

Incompressible Navier-Stokes Equation

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

- Suppose we simulate a fluid with a <u>particle system</u>
- Each particle has a mass m, a volume V, and a velocity $\vec{u}$
- To simulate the fluid, we use the same approach as before !
- For each particle
  - Finding all forces applied F_i
  - Finding the acceleration $\sum F_i = ma$
  - From the acceleration, find the velocity $a \equiv \frac{D\vec{u}}{Dt}$

Forces and Material Derivative

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

- Note that I used a strange notation for the derivative $a \equiv \frac{D\vec{u}}{Dt}$
- This is called the **material derivative**
  - For the moment, let's think of it as the normal derivative
- Our newton laws becomes $F_{tot} = ma = m\frac{D\vec{u}}{Dt}$
- What are the forces applied to a fluid?

Forces and Material Derivative

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

- Note that I used a strange notation for the derivative $a \equiv \frac{D\vec{u}}{Dt}$
- This is called the **material derivative**
  - For the moment, let's think of it as the normal derivative
- Our newton laws becomes $F_{tot} = ma = m\frac{D\vec{u}}{Dt}$
- What are the forces applied to a fluid?
- The forces applied to our particle include of course the gravity g

$$F_g = mg$$

- What else ?

Forces due to Pressure

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$

$$\nabla \cdot \vec{u} = 0$$

- It is interesting to know what forces are applied to particles by <u>other particles</u>
- For example the **pressure** in the fluid
- Pressure arise from microscopic collision between particles
  - Constant pressure in every direction = particles do not add forces to other particles
  - We observe an effect of pressure on particles when there is an imbalance
- The pressure is whathever it takes to keep the fluid at constant volume

Forces due to Pressure

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

- The gradient of the pressure represents how pressure changes in space
- A fluid accelerates from regions of high pressure to regions of low pressure
- The simplest way to take into account the pressure as a force, is to take its gradient: $-\nabla p$
- We want to integrate it over the whole fluid to have the total pressure
- A simpler approach is to multiply it by the volume:

$$F_p = -\nabla p . V$$

Forces due to Viscosity

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

- Another important force is the **viscosity** of the fluid
- A viscouous fluid tries to resist deforming (think honey)
- Intuitively, its a force that tries to make the particles move at the average velocity of nearby particles

- The differential operator that measures how far a quantity is from the average around is the laplacian $\nabla \cdot \nabla$ of our velocity $\vec{u}$

- We use the **dynamic viscosity coefficient** μ (resistance to deformationsdue to velocity differences)
- and multiply by the volume to have a force:

$$F_v = V \mu \nabla \cdot \nabla \vec{u}$$

All our forces together

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

- Putting all our forces together we have

$$\sum F_i = m \frac{D\vec{u}}{Dt} \Leftrightarrow F_g + F_P + F_v = m \frac{D\vec{u}}{Dt}$$

- i.e,

$$m \frac{D\vec{u}}{Dt} = m\vec{g} - V\nabla \mathrm{p} + \mathrm{V}\mu\nabla \cdot \nabla \vec{u}$$

- We make lot's of errors when we approximate a fluid using particles,
- We can take the limit of their mass and volumes to tend toward 0

- This limit is called the **continuum model**, it is independent from the number of particles in our system

All our forces together

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

$$m \frac{D\vec{u}}{Dt} = m\vec{g} - V\nabla \mathrm{p} + \mathrm{V}\mu\nabla \cdot \nabla \vec{u}$$

- However, taking the limit makes the simulation too computationally expensive and generates number close to zero, which is not stable to perform math operations on a computer !
- What to do instead?

All our forces together

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

$$m \frac{D\vec{u}}{Dt} = m\vec{g} - V\nabla \mathrm{p} + V\mu\nabla \cdot \nabla \vec{u}$$

- However, taking the limit makes the simulation too computationally expensive and generates number close to zero, which is not stable to perform math operations on a computer !
- What to do instead?
- We can divide by the volume of the fluid !

$$\frac{m}{V} \frac{D\vec{u}}{Dt} = \frac{m}{V} \vec{g} - \frac{V}{V} \nabla \mathrm{p} + \frac{\mathrm{V}}{\mathrm{V}} \mu \nabla \cdot \nabla \vec{u}$$

- Remembering that a density is $\rho = \frac{m}{V}$, we have

$$\rho \frac{D\vec{u}}{Dt} = \rho \vec{g} - \nabla \mathrm{p} + \mu \nabla \cdot \nabla \vec{u}$$

All our forces together

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

$$m \frac{D\vec{u}}{Dt} = m\vec{g} - V\nabla \mathrm{p} + V\mu \nabla \cdot \nabla \vec{u}$$

- However, taking the limit makes the simulation too computationally expensive and generates number close to zero, which is not stable to perform math operations on a computer !
- What to do instead?
- We can divide by the volume of the fluid !

$$\frac{m}{V} \frac{D\vec{u}}{Dt} = \frac{m}{V} \vec{g} - \frac{V}{V} \nabla \mathrm{p} + \frac{V}{V} \mu \nabla \cdot \nabla \vec{u}$$

- Remembering that a density is $\rho = \frac{m}{V}$, we have

$$\rho \frac{D\vec{u}}{Dt} = \rho \vec{g} - \nabla \mathrm{p} + \mu \nabla \cdot \nabla \vec{u}$$

- Looking familiar?

All our forces together

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

$$\rho \frac{D\vec{u}}{Dt} = \rho \vec{g} - \nabla \mathrm{p} + \mu \nabla \cdot \nabla \vec{u}$$

- Dividing by the density leads to:

$$\frac{D\vec{u}}{Dt} = \vec{g} - \frac{1}{\rho} \nabla \mathrm{p} + \frac{\mu}{\rho} \nabla \cdot \nabla \vec{u}$$

- And rearranging a bit

$$\frac{D\vec{u}}{Dt} + \frac{1}{\rho} \nabla p = \vec{g} + \frac{\mu}{\rho} \nabla \cdot \nabla \vec{u}$$

- We finally define the kinematic viscosity $\nu = \frac{\mu}{\rho}$

$$\frac{D\vec{u}}{Dt} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$

- We are almost there ! There is still the mistery of the material derivative !

Material derivative

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$

$$\nabla \cdot \vec{u} = 0$$

$$\frac{D\vec{u}}{Dt} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$

- We are almost there ! There is still the mistery of the material derivative !
- But first, we need to understand the **Lagrangian** and **Eulerian** viewpoints !
- When we think about a fluid, there are 2 ways of tracking its motion
  - The Lagrangian and the Eulerian viewpoints
- Think of them as two different ways of explaining the same situation

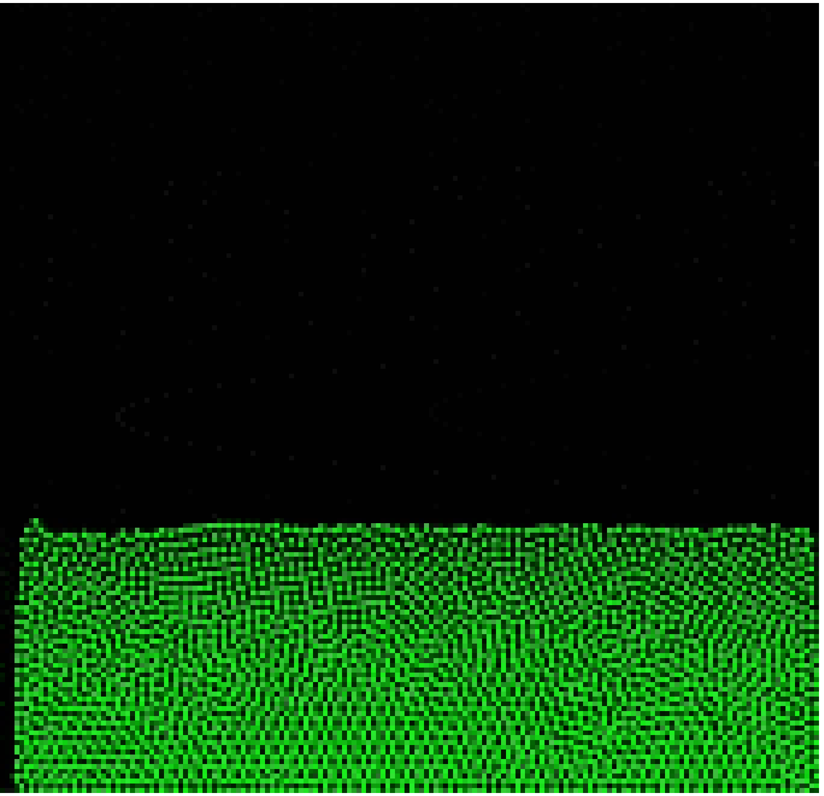
Lagrangian Viewpoint

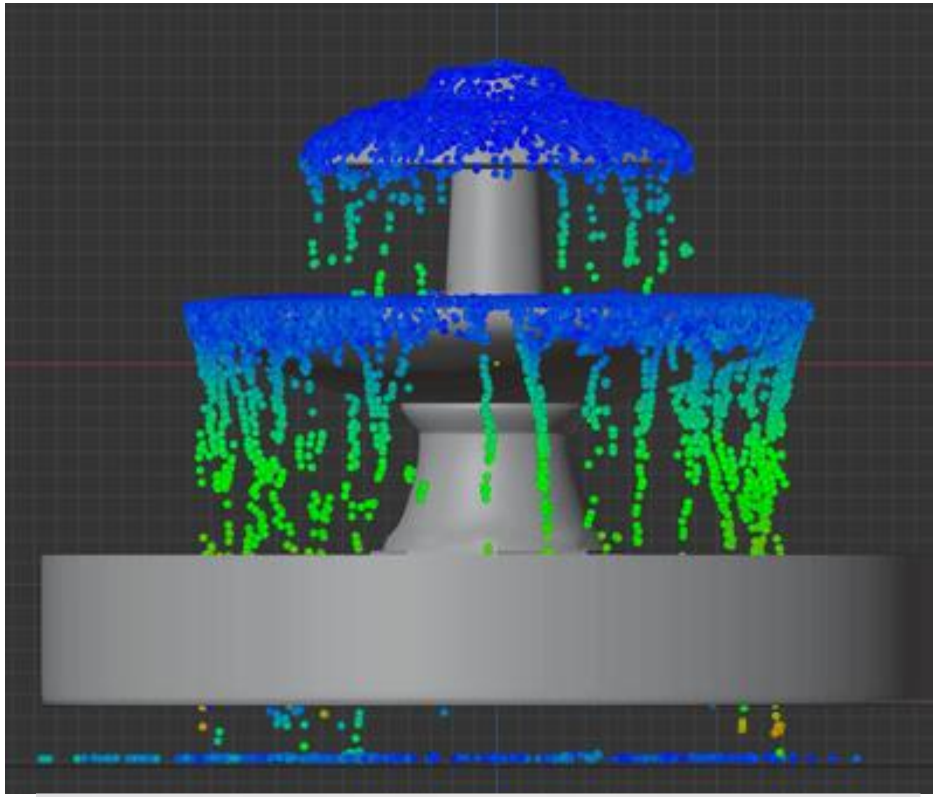
- The **Lagrangian viewpoint** (You are mostly familiar it):
  - Consider a fluid like a particle system.
  - Each particle has a position $\vec{x}$ and a velocity $\vec{u}$.
  - Solids are almost always simulated in a Lagrangian way

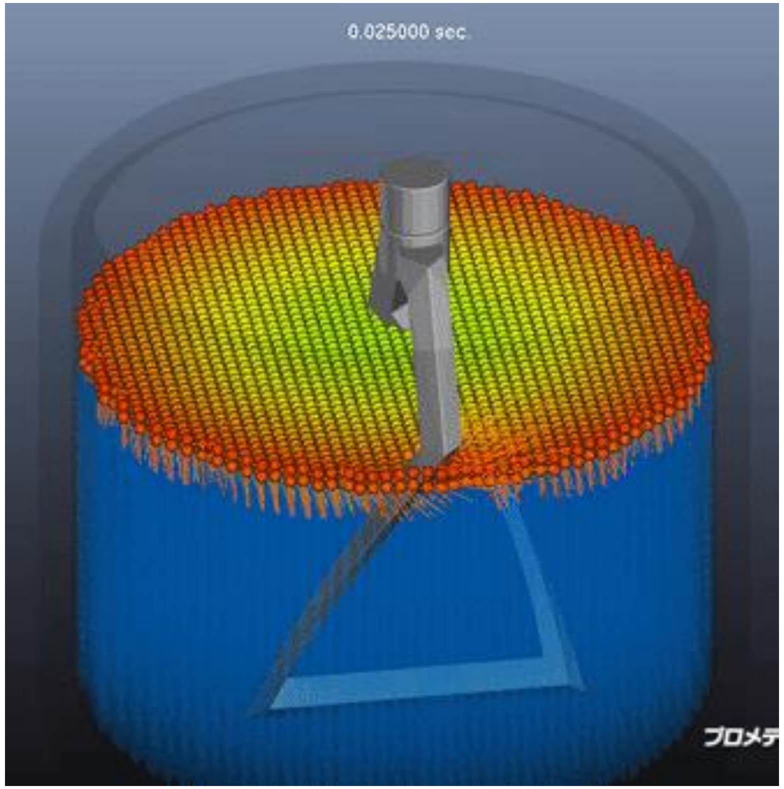
- Each particle represents a blob of the liquid, not individual particles as it would be impossible to simulate.

- To each blob, we can attach quantities such as pressure, temperature, density, etc.
  - Those quantities represent the internal status of the blob

Examples of the Lagrangian Viewpoint

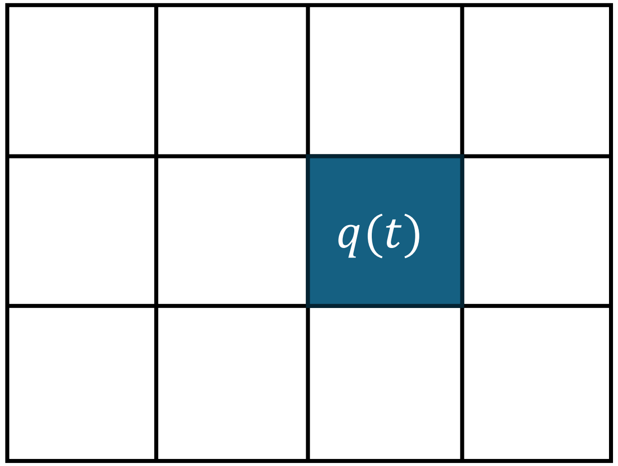




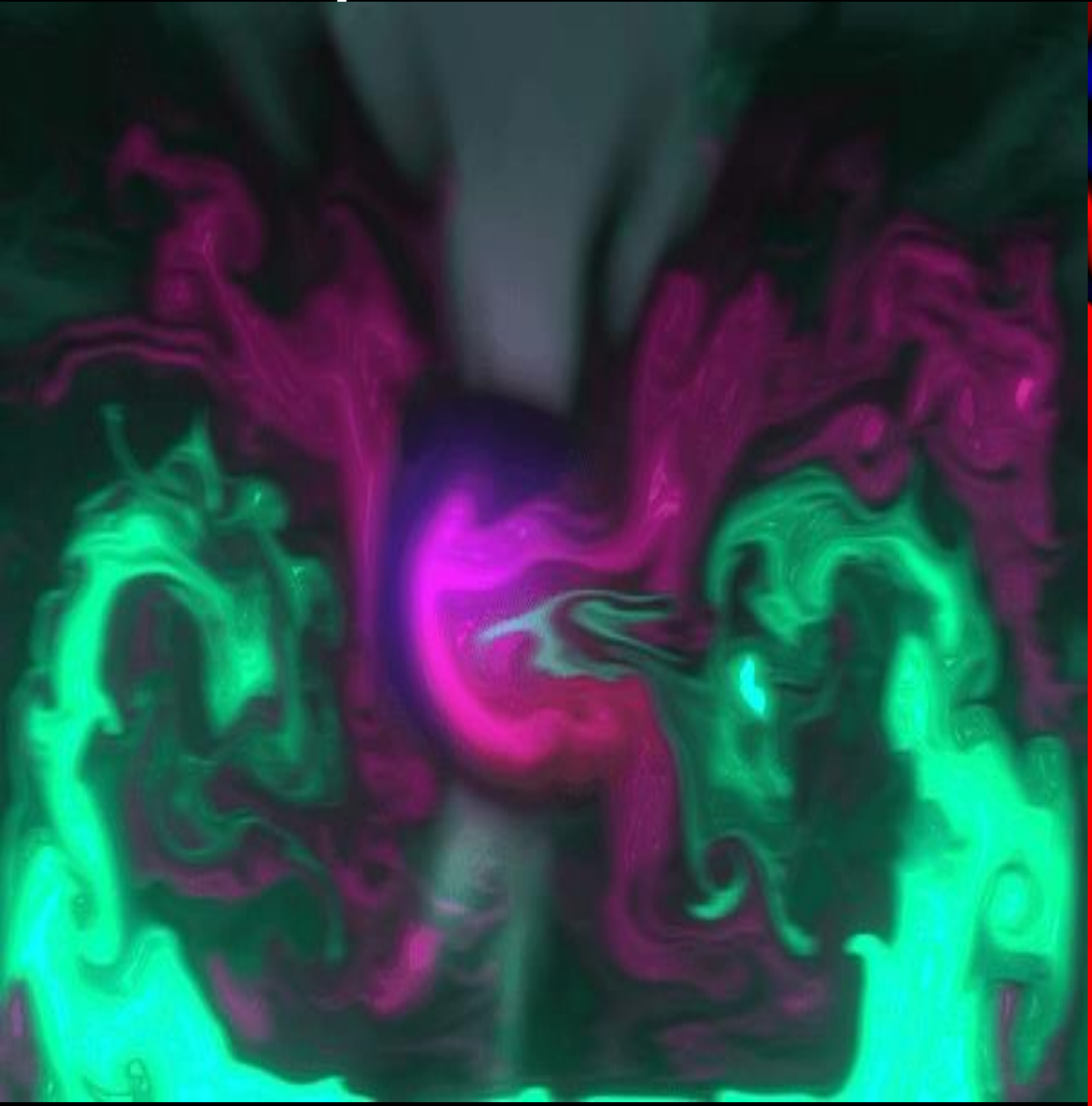


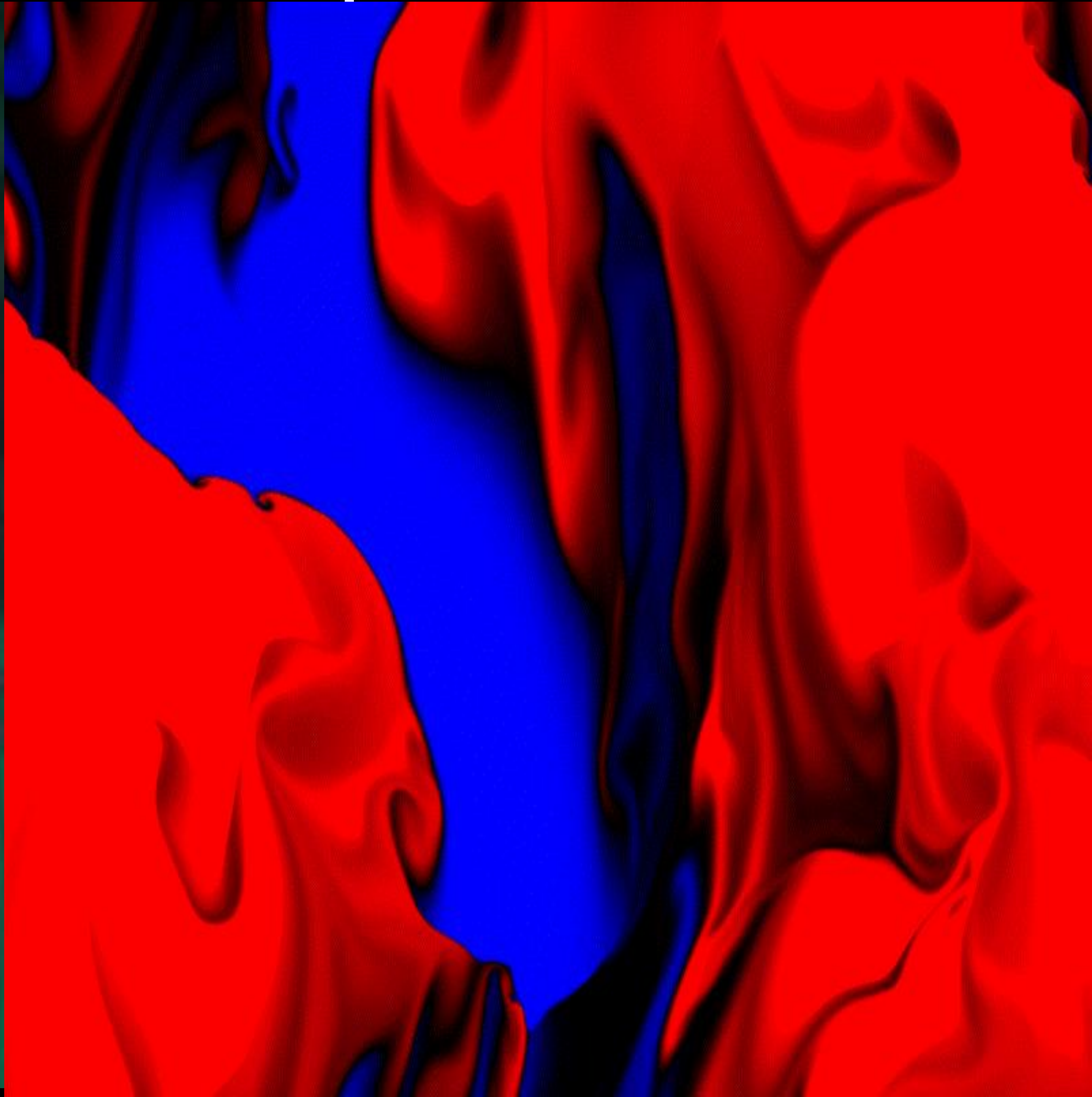
Eulerian Viewpoint

- **Eulerian viewpoint** (most practical for fluid simulations):
  - Instead of tracking each particle, we look at <u>fixed points</u> in space <u>in a grid</u>
  - For each points, we observe how quantities (e.g: density, temperature) evolve with time
  - The fluid is flowing past those points, contributing to the value observed
    - E.g.: If a warm fluid passes through a point, then a cold fluid passes through the same point, the temperature at the point will change. EVEN if the particles in the fluids didn't change !
  - In addition, other fluid variables can be changing in the fluid, contributing in other ways to the fixed point
    - E.g.: If globally a fluid cools off, each point in a grid will decrease the temperature value

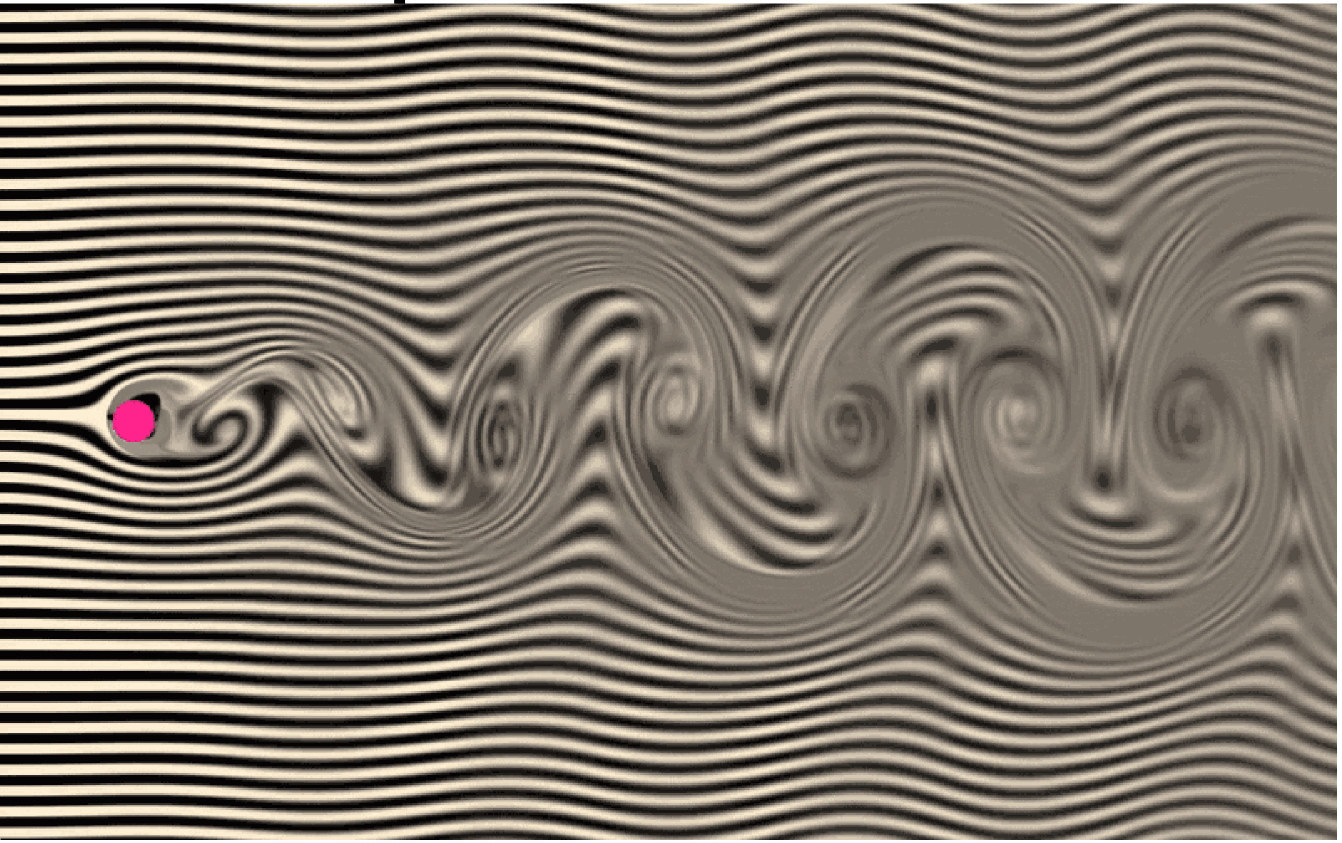


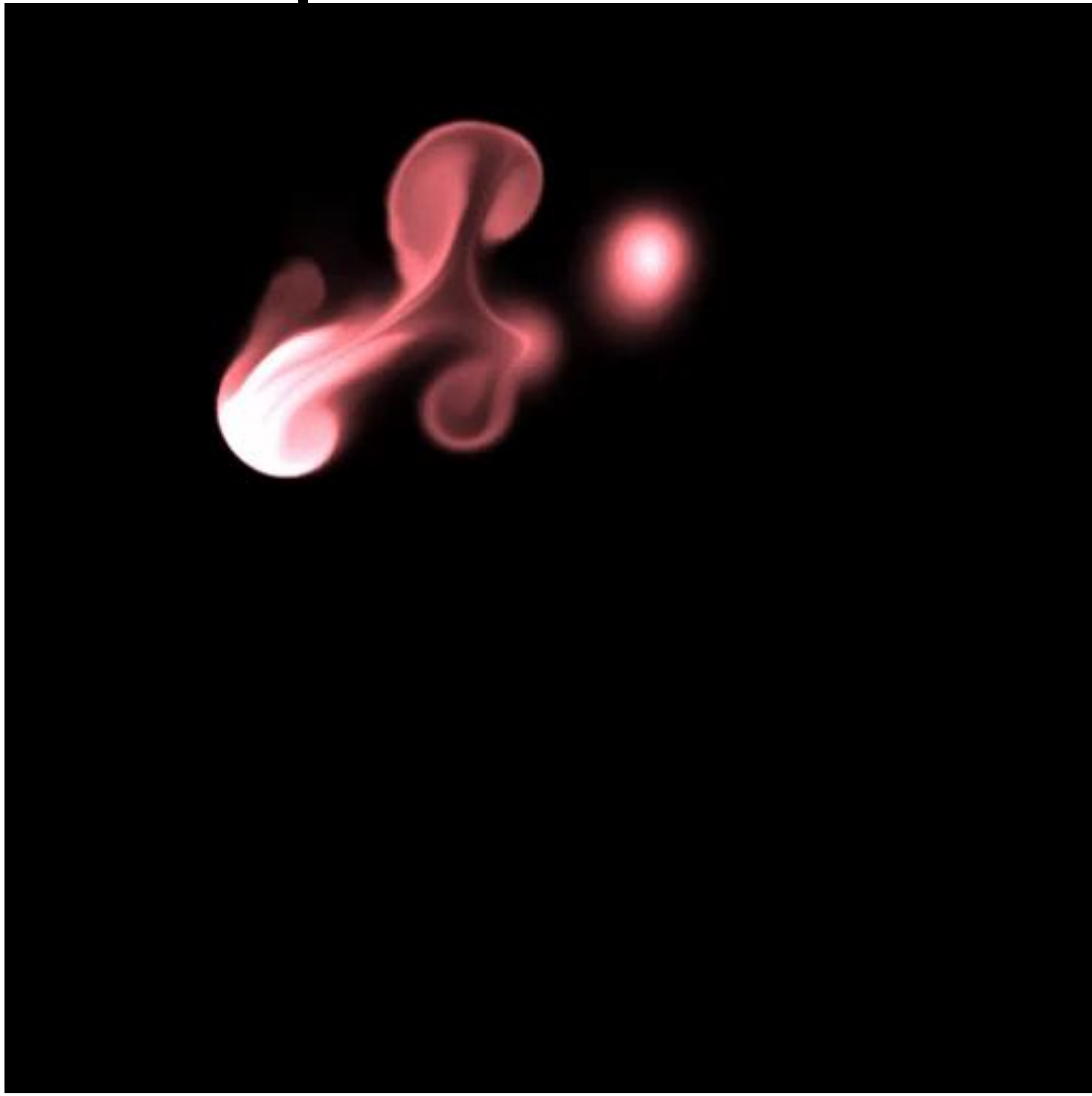
Examples of the Eulerian Viewpoint

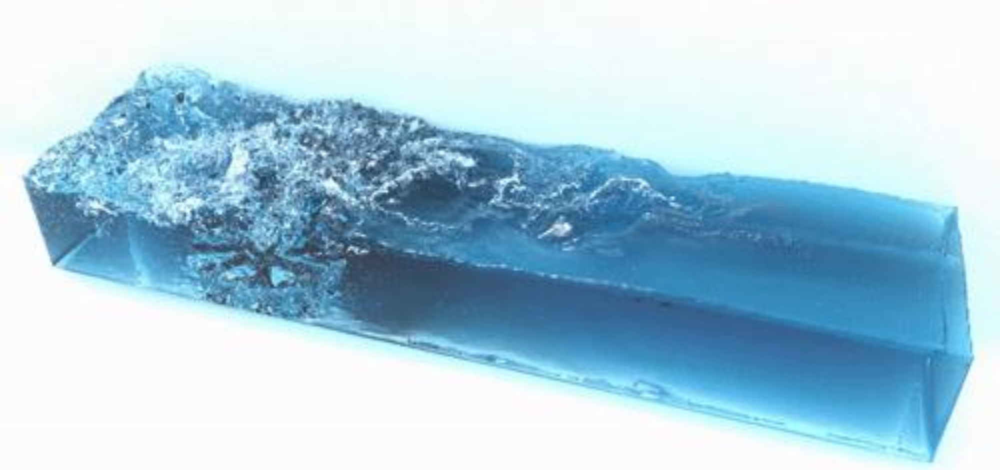




Examples of the Eulerian Viewpoint







Lagrangian and Eulerian Viewpoints

- Imagine there are two photographers, Mr. **Euler** and Mr. **Lagrange**, in a parade.
- Lagrange moves along the parade, hopping onto a float and traveling down the streets with it,
  capturing the parade from within
- Euler sets ups a tripod on the sidewalk and stays in on the spot,
  capturing photos as the parade passes by.

Lagrangian and Eulerian Viewpoints

- How their view differ?
- **Lagrange** perspective:
  - Lagrange is traveling with a particular flow. He captures pictures from within the parade itself, staying close to the same participants as they move along the route.
  - It shows how the view changes as the parade moves through different parts of the city. He will observe changes in the background scenery (buildings, trees, crowds,…) as he moves, but will stay focused on the same group in the parade.
- **Euler** perspective:
  - Euler captures the parade from a fixed spot on the sidewalk. Every few moments, a new dancer, person, animal passes by, and he photograph them from his stationary position.
  - He observes how the parade changes over time at a specific point. For example, he may see a group with red balloon followed by a group of dancers, and then a band with trumpets.
  - His photos document the sequence of participants passing by, but not of the entire parade.

Lagrangian approach

- **Lagrangian** approach:
  - **Focus**: Tracks individual particles or fluid elements.
  - **Applications**:
    - **Environmental Science**: Monitoring pollutant movement in water bodies.
    - **Biomechanics**: Analyzing blood flow and nutriment transport in biological systems.
    - **Material Science**: Studying aerosols, emulsions, and colloidal suspensions
  - **Ideal For**: Fluid mixing, particle dispersion, wave propagation
- **Eulerian** approach:
  - **Focus**: Observes the overall behavior of the fluid at fixed points in space
  - **Applications**:
    - **Meteorology**: Analyzing weather patterns and atmospheric phenomena
    - **Oceanography**: Understanding currents, wave dynamics, and oceanic heat distribution.
    - **Engineering**: CFD simulations for airflow optimization, combustion studies.
  - **Ideal For**: Flow patterns, turbulence, pressure distribution

Connecting Lagrangian and Eulerian Perspectives

- The **material derivative,** $\frac{Dq}{Dt}$, links:
  - **Lagrangian** Perspective: Following individual particles to observe how a quantity q, (e.g., temperature) changes along their path.
    - The particles have positions $\vec{x}$ and velocities $\vec{u}$
    - Each particle has a quantity, q, (e.g., temperature, density) that may change as the particle moves
  - **Eulerian** Perspective: Observing changes in q at a fixed location in space.
    - Observes $q(t, \vec{x})$ at a fixed location, showing how q evolves at that spot over time.

- This concept captures changes in q as it flows within a fluid.

Demonstrating the Material Derivative

- The material derivative combines Eulerian and Lagrangian changes:
- Starting Point (Lagrangian):
  - Consider $q(t, \vec{x})$, the value of q for a moving particle at position $\vec{x}(t)$
- Taking the derivative with respect to time:
  - We use the chain rule to differentiate q along the particle's path:

$$\frac{d}{dt} q\big(t, \vec{x}(t)\big) = \frac{\partial q}{\partial t} + \nabla q \cdot \frac{d\vec{x}}{dt}$$

- Substitute Particle Velocity:
  - Since $\frac{d\vec{x}}{dt} = \vec{u}$ (the particle velocity),

$$\frac{Dq}{Dt} \equiv \frac{\partial q}{\partial t} + \nabla q \cdot \vec{u}$$

Understanding the terms

$$\frac{Dq}{Dt} \equiv \frac{\partial q}{\partial t} + \nabla q \cdot \vec{u}$$

- **Local Rate of Change** $\left(\frac{\partial q}{\partial t}\right)$: The change of q at a fixed spatial location (Eulerian)
- **Convective Rate of Change** $(\nabla q \cdot \vec{u})$: Adjusts for changes as particles carrying q flow with velocity $\vec{u}$ (Lagrangian).
  - E.g., the temperature changing because hot air is being replaced by cold air. Not because the temperature of any molecule is changing.

Why Both Terms Matter?

- Imagine q is temperature in a moving fluid:
  - $\frac{\partial q}{\partial t}$ alone would show how the temperature varies at a fixed spot, like reading a termometer
  - $\nabla q \cdot \vec{u}$ corrects for temperature changes due to warmer or cooler fluid moving through that point

- Takeaway:
  - $\frac{Dq}{Dt}$ (material derivative) provides a Lagrangian perspective: It tells us how q changes over time for a moving particle.
  - The material derivative connects Lagrangian and Eulerian views, capturing the complete rate of change in moving fluids.

Example of Material Derivative Application

- As an example, suppose the quantity q is moving around, but isn't changing in the Lagrangian viewpoint.
- Then, an **advection equation** is just one that uses the material derivative and it sets it to zero.

$$\frac{Dq}{Dt} = 0$$

Lagrangian

- I.e.,

$$\frac{\partial q}{\partial t} + \vec{u} \cdot \nabla q = 0$$

Eulerian

- We have pure advection, i.e, no internal changes
- When $\frac{Dq}{Dt} = 0$, q only changes because of its movement with the fluid (advection), not because it's evolving internally !

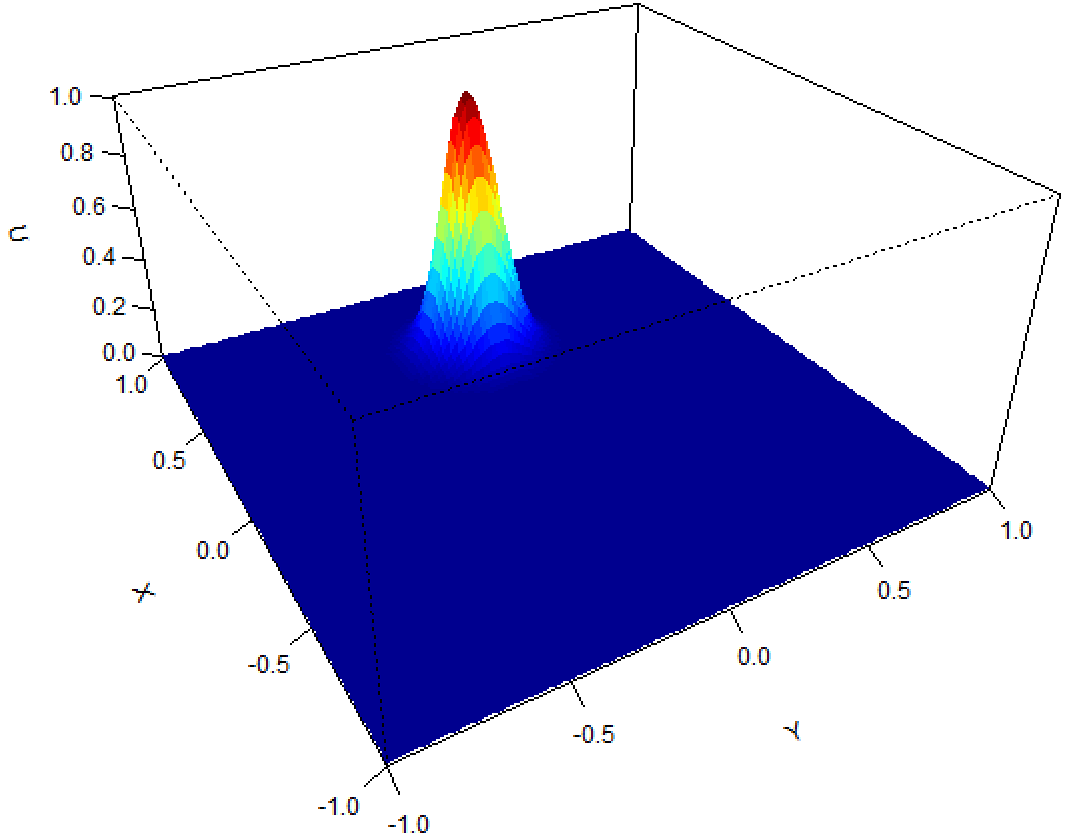
Practical example of advection

- Suppose we have a quantity temperature $T(x) = 10x$
  - Cold at the origin, and 100 degrees at $x = 10$
- Say a steady wind of speed c is blowing: $\vec{u} = c$

- We assume the particles' temperature in the air isn't changing, just moving.
- In the lagrangian viewpoint: $\frac{DT}{Dt} = 0$
- By expanding we have: $\frac{\partial T}{\partial t} + \vec{u} \cdot \nabla T = 0 \Leftrightarrow \frac{\partial T}{\partial t} + 10c = 0 \Leftrightarrow \frac{\partial T}{\partial t} = -10c$
- So, in the Eulerian point of view, we observe the drop in temperature at a rate $-10c$. If the wind has stopped, $c = 0$, nothing changes. If the wind blows at $c = 1$ to the right, the temperature at a fixed point will drop at a rate of -10

Practical example of advection

- Even though the Lagrangian derivative is zero, the Eulerian derivative can be anything depending on how fast and in what direction the flow is moving.



The Incompressibility Condition

$$\frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{u} + \frac{1}{\rho} \nabla p = \vec{g} + \nu \nabla \cdot \nabla \vec{u}$$
$$\nabla \cdot \vec{u} = 0$$

- There is a last term we did not describe yet: $\nabla \cdot \vec{u} = 0$
- This term is the incompressibility condition

- Liquids change their volume over time, but usually, not very much !
- In general we treat all fluids as incompressible in animation.

- We have $\frac{d}{dt} Volume(\Omega) = \iint_{\partial\Omega} \vec{u} \cdot \vec{n} = 0$
  - I.e., the velocity at the borders of the surface is zero !

- We use the divergence theorem to transform this integral into a volume integral: $\iint_{\partial\Omega} \vec{u} \cdot \vec{n} = \iiint_{\Omega} \nabla \cdot \vec{u} = 0$, this is true iff $\nabla \cdot \vec{u}$, our condition.

Border conditions

- We focused on what happens inside the fluid
- But what happens at the borders is also essential

- We use two kind of borders:
  - **Solid walls**: We set that the velocity of the fluid is zero at the border $n \cdot \vec{u} = 0$
    - Note: If the wall is moving, we need to match the speed of the wall instead $n \cdot \vec{u} = n \cdot \vec{u}_{wall}$
  - **Free Surfaces**: We do not want to simulate air, we set the air pressure to a convenient value, eg: $p = 0$, and we do not control the velocity of the fluid

Fluid Animations

- Now you know the basics for fluids animations !

- To do an Euler animation:
  - You define a grid
  - You compute the Navier-Stockes equation at each point
  - You take into account border conditions

What I Expect for the exam

- After this lesson I expect that:
  - You know what a particle system is
  - You know how to animate a particle system
  - You understand the tradeoffs and derivations of
      - Forward, Backward and, Symplectic Euler
  - You know how springs work and how to animate hairs and clothes
  - You have an idea of what are the tradeoffs for collisions
  - You can explain in your words the link with rigid bodies
  - You have an idea of how fluids work, in particular the Eulerian and Lagrangian point of view
  - You can explain, INTUITIVELY, what the Navier-Stokes equation says
- I DO NOT EXPECT
  - That you are able to re-derive the formulas for rigid bodies
  - That you are able to re-derive the formulas for fluids

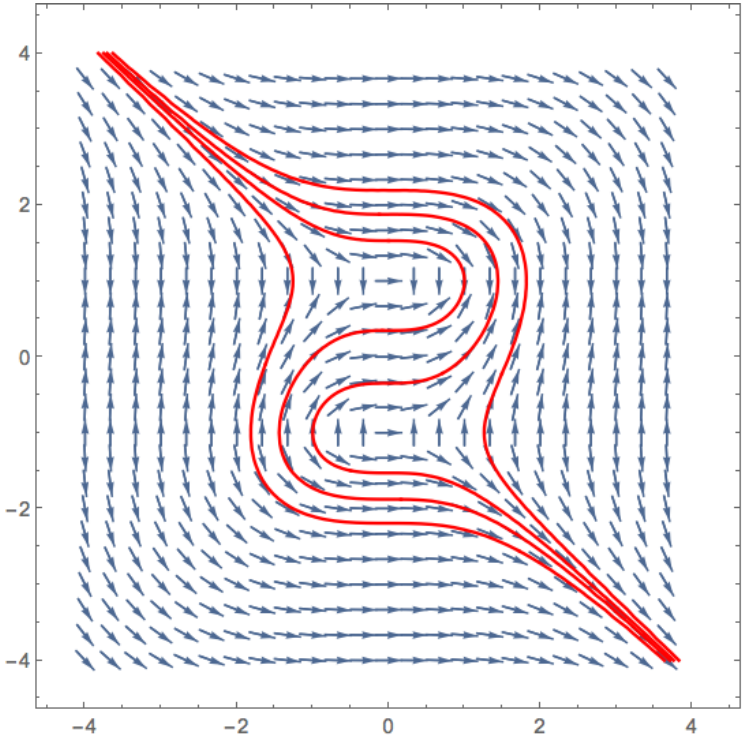
Next time

Appendix – Newton Law of Motion

$$F = ma = m\ddot{x}$$

- This is a differential equation in x
- A differential equation has a solution a vector field, where each arrow in the vector field represent a good solution to our equation.
- By specifying initial values, eg $x(5) = 3$, we select which path we follow in our field.

- When we integrate, we try to stay on the same path, all the integration techniques try to refine our approximation of the true path.



Appendix: Integrations !

- Forward, Backward, Symplectic Euler, Midpoint are only a few techniques of integration.
- When it is very important to have a good integral, for example:
  - Highly oscillating system
  - Orbitals of planets
- It is better to use more advanced methods, in particular the Runge-Kutta 4 Method.
- It will be briefly described ~~in the following slides~~, along with the Verlet integration

Verlet Integration

- When we integrate, we never used the fact that we already computed the previous position !
- Using this in our estimation could lead to a better approximation

- Let's use Taylor expansion for the following position and the previous one
  - $x_{t+\Delta t} = x(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2 + O(\Delta t^3)$
  - $x_{t-\Delta t} = x(t) - v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2 + O\left(\Delta t^3\right)$
- Adding them together:
  - $x_{t+\Delta t} + x_{t-\Delta t} = 2x(t) + a(t)\Delta \mathrm{t}^2 + \mathrm{O}(\Delta t^3)$
  - Therefore:
    - $x_{t+\Delta t} \approx -x_{t-\Delta t} + 2x(t) + a(t)\Delta \mathrm{t}^2$

- We do not need anymore the velocity !

Appendix: Eulerian vs Lagrangian viewpoints

- Here are some more examples to work yourself on your comprehension:
  - **Weather Observation**: Mr. Euler stands at a weather station, noting wind and temperature changes at one spot, while Mr. Lagrange follows a weather balloon through the sky, tracking conditions along its path.
  - **Traffic Flow**: Mr. Euler watches cars passing a traffic light, noting speeds and patterns, while Mr. Lagrange follows a specific car on the highway, observing speed changes over its journey.
  - **Forest Fire Spread**: Mr. Euler stands at a fire lookout, observing how flames reach his location, while Mr. Lagrange follows a burning ember, noting how it spreads across the forest.
  - **Sound Waves in a Concert Hall**: Mr. Euler remains at the back of the hall, hearing changes in sound volume and quality, while Mr. Lagrange moves with a sound wave, observing how it interacts with walls and objects.
  - **Delivery Route Tracking**: Mr. Euler watches packages arrive at a distribution hub, tracking delivery times at a fixed point, while Mr. Lagrange rides along with a delivery truck, noting route changes and delays.

Appendix: Eulerian vs Lagrangian viewpoints

- Here are some more examples to work yourself on your comprehension:
  - **River Water Quality**: Mr. Euler samples water quality from a dock, noting changes over time at one location, while Mr. Lagrange floats downriver, tracking how quality changes along the journey.
  - **Bee Foraging Patterns**: Mr. Euler observes bees visiting flowers in a garden, noting frequencies and patterns at his spot, while Mr. Lagrange follows a single bee, tracking its path and visited flowers.
  - **Elevator Traffic in a Building**: Mr. Euler stays on the ground floor, observing how often the elevator arrives, while Mr. Lagrange rides the elevator, noting how crowded it gets on different floors.
  - **Subway Station Monitoring**: Mr. Euler stays at a single station, tracking passenger flow and train arrivals, while Mr. Lagrange boards a train, observing how crowding changes from station to station.
  - **Bird Migration**: Mr. Euler waits at a known migration point, counting birds as they pass, while Mr. Lagrange flies with a bird, noting the journey's conditions and rest stops

Appendix: Applying the material derivative

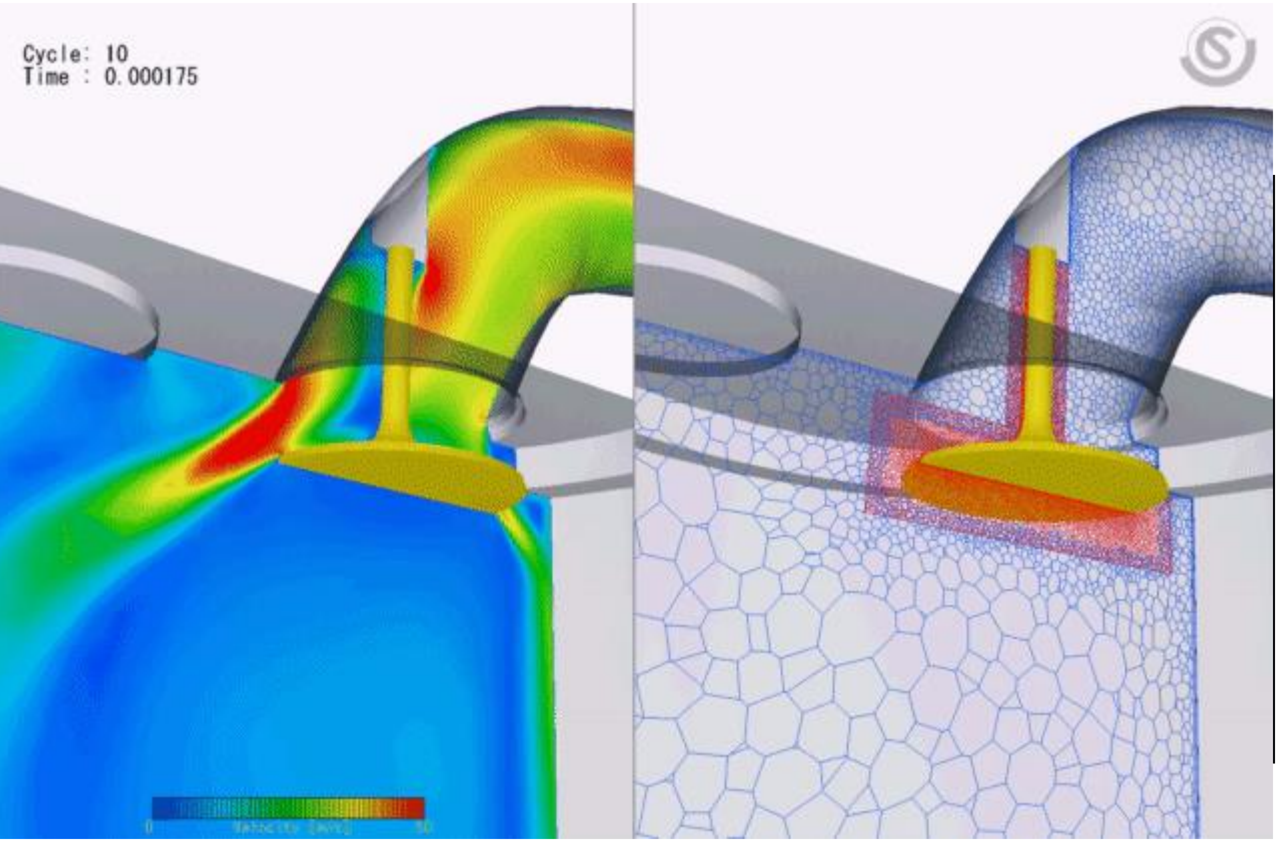
- The material derivative can be applied to vector quantities, like RGB colors or $\vec{u}$ itself.
- We treat each component separately:
- Let $\vec{\mathrm{u}} = (u, v, w)$, then

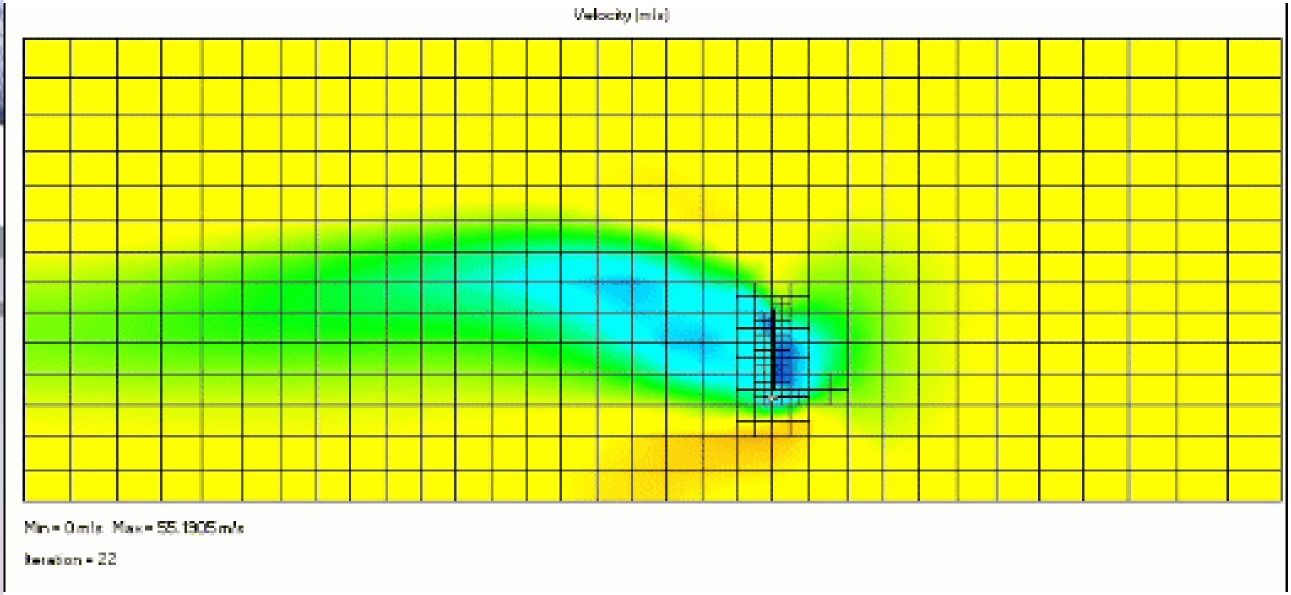
$$\frac{D\vec{u}}{Dt} = \begin{bmatrix} \frac{Du}{Dt} \\ \frac{Dv}{Dt} \\ \frac{Dw}{Dt} \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial t} + \vec{u} \cdot \nabla u \\ \frac{\partial v}{\partial t} + \vec{u} \cdot \nabla v \\ \frac{\partial w}{\partial t} + \vec{u} \cdot \nabla w \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} \\ \frac{\partial u}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z} \\ \frac{\partial u}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} \end{bmatrix} = \frac{\partial \vec{u}}{\partial t} + \vec{u} \cdot \nabla \vec{\mathrm{u}}$$

- The notation $\vec{u} \cdot \nabla \vec{u}$, make sense only componentwise.

Appendix: Eulerian Viewpoint - Grids

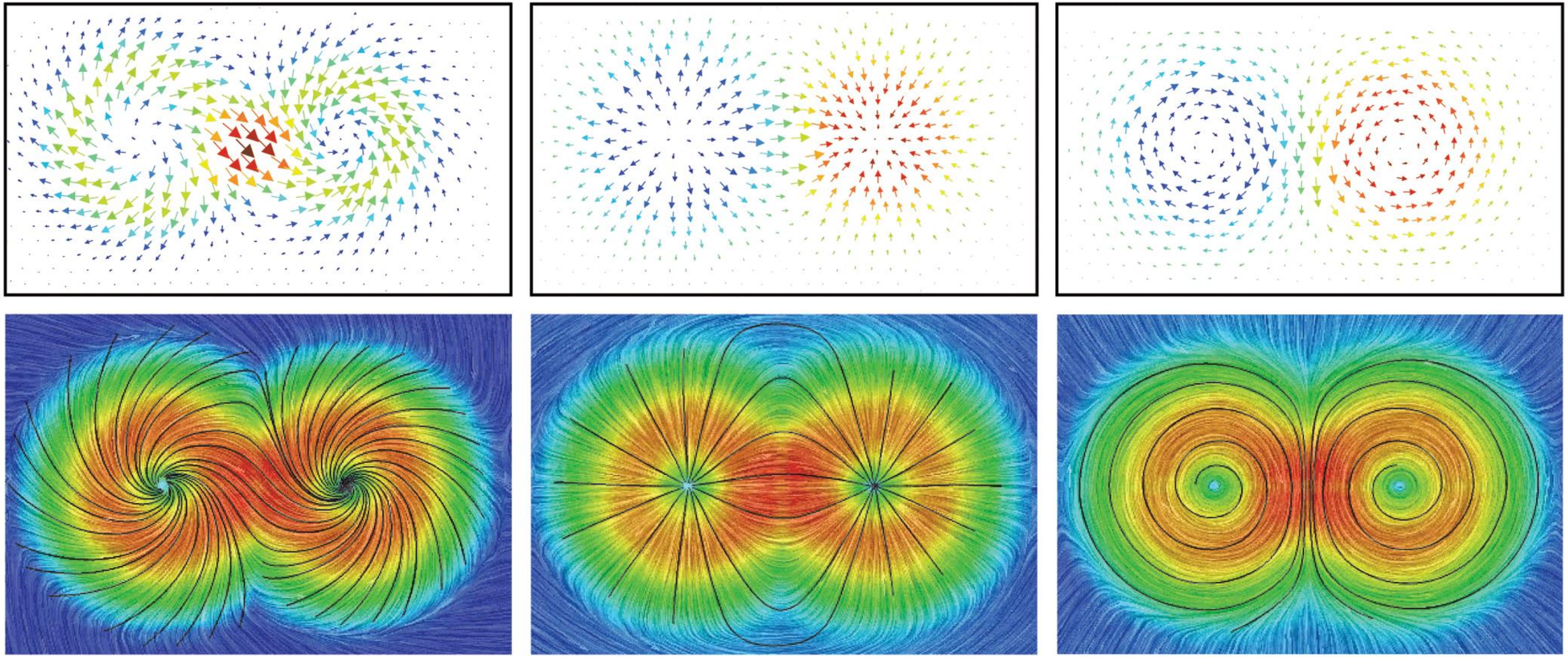
- Grids do not have to be uniform, they can be adaptative !





Appendix: Helmholtz Decomposition

- The Helmholtz decomposition states that any sufficiently smooth vector field can be expressed as the sum of a irrotational field and a divergence free field



Appendix: Linear Momentum

3blue1brown

- Momentum is the property of the object that makes it easier or harder to stop, eg, a very heavy mass vs a light one

Appendix: Inertia Tensor - Proof

- The angular momentum L of a single particle is:

$$L = r \times p = r \times mv$$

- For rotational motion:

$$v = \omega \times r$$

- If we substitute we have

$$L = mr \times (\omega \times r)$$

- By using the triple cross product identity:

$$r \times (\omega \times r) = (r \cdot r)\omega - (r \cdot \omega)r$$

- Thus, for a single particle

$$L = m[(r \cdot r)\omega - (r \cdot \omega)r]$$

Appendix: Inertia Tensor - Proof

- For a single particle

$$L = m[(r \cdot r)\omega - (r \cdot \omega)r]$$

- For a system of N particles:

$$L = \sum_{i}^{N} m_i[(r_i \cdot r_i)\omega - (r_i \cdot \omega)r_i]$$

$$L = \sum_{i}^{N} m_i[(r_{ix}r_{ix} + r_{iy}r_{iy} + r_{iz}r_{iz})\omega - (r_{ix}\omega_x + r_{iy}\omega_y + r_{iz}\omega_z)r_i]$$

$$L = \sum_{i}^{N} m_i\left[\begin{pmatrix} (r_{ix}r_{ix} + r_{iy}r_{iy} + r_{iz}r_{iz})\omega_x \\ (r_{ix}r_{ix} + r_{iy}r_{iy} + r_{iz}r_{iz})\omega_y \\ (r_{ix}r_{ix} + r_{iy}r_{iy} + r_{iz}r_{iz})\omega_z \end{pmatrix} - \begin{pmatrix} (r_{ix}\omega_x + r_{iy}\omega_y + r_{iz}\omega_z)r_{ix} \\ (r_{ix}\omega_x + r_{iy}\omega_y + r_{iz}\omega_z)r_{iy} \\ (r_{ix}\omega_x + r_{iy}\omega_y + r_{iz}\omega_z)r_{iz} \end{pmatrix}\right]$$

Appendix: Inertia Tensor - Proof

$$L = \sum_i^N m_i \left[\begin{pmatrix} (r_{ix}r_{ix} + r_{iy}r_{iy} + r_{iz}r_{iz})\omega_x \\ (r_{ix}r_{ix} + r_{iy}r_{iy} + r_{iz}r_{iz})\omega_y \\ (r_{ix}r_{ix} + r_{iy}r_{iy} + r_{iz}r_{iz})\omega_z \end{pmatrix} - \begin{pmatrix} (r_{ix}\omega_x + r_{iy}\omega_y + r_{iz}\omega_z) r_{ix} \\ (r_{ix}\omega_x + r_{iy}\omega_y + r_{iz}\omega_z) r_{iy} \\ (r_{ix}\omega_x + r_{iy}\omega_y + r_{iz}\omega_z) r_{iz} \end{pmatrix} \right]$$

Let's keep only the k-th component of L $(k = 1, 2, 3)$

$$L_k = \sum_i^N m_i \left[(r_{ix}r_{ix} + r_{iy}r_{iy} + r_{iz}r_{iz})\omega_k - (r_{ix}\omega_x + r_{iy}\omega_y + r_{iz}\omega_z) r_{ik} \right]$$

$$L_k = \sum_i^N m_i \left[\sum_j^3 r_{ij}r_{ij}\,\omega_k - \sum_j^3 r_{ij}\omega_j\, r_{ik} \right]$$

But $\omega_k = \sum_j^3 \omega_j \delta_{kj}$

Appendix: Inertia Tensor - Proof

$$L_k = \sum_i^N m_i \left[\sum_j^3 r_{ij} r_{ij}\, \omega_k - \sum_j^3 r_{ij} \omega_j\, r_{ik} \right]$$

But $\omega_k = \sum_j^3 \omega_j \delta_{kj}$

$$L_k = \sum_i^N m_i \left[\sum_j^3 r_{ij} r_{ij}\, \omega_j \delta_{kj} - \sum_j^3 r_{ij} \omega_j\, r_{ik} \right]$$

Regrouping the sum over j

$$L_k = \sum_i^N m_i \left[\sum_j^3 r_{ij} r_{ij}\, \omega_j \delta_{kj} - r_{ij} \omega_j r_{ik} \right]$$

Appendix: Inertia Tensor - Proof

$$L_k = \sum_i^N m_i \left[\sum_j^3 r_{ij} r_{ij}\, \omega_j \delta_{kj} - r_{ij} \omega_j r_{ik} \right]$$

$$L_k = \sum_i^N m_i \left[\sum_j^3 [r_{ij} r_{ij} \delta_{kj} - r_{ij} r_{ik}]\, \omega_j \right]$$

$$L_k = \sum_j^3 \omega_j \sum_i^N m_i [r_{ij} r_{ij} \delta_{kj} - r_{ij} r_{ik}]$$

Appendix: Inertia Tensor - Proof

$$L_k = \sum_j^3 \omega_j \sum_i^N m_i\left[r_{ij}r_{ij}\delta_{kj} - r_{ij}r_{ik}\right]$$

$$L_k = \omega_x \sum_i^N m_i[r_{ix}r_{ix}\delta_{kx} - r_{ix}r_{ik}] + \omega_y \sum_i^N m_i\left[r_{iy}r_{iy}\delta_{ky} - r_{iy}r_{ik}\right] + \omega_z \sum_i^N m_i[r_{iz}r_{iz}\delta_{kz} - r_{iz}r_{ik}]$$

$$L_k = (\sum_i^N m_i[r_{ix}r_{ix}\delta_{kx} - r_{ix}r_{ik}] \quad \sum_i^N m_i\left[r_{iy}r_{iy}\delta_{ky} - r_{iy}r_{ik}\right] \quad \sum_i^N m_i[r_{iz}r_{iz}\delta_{kz} - r_{iz}r_{ik}])\omega$$

Appendix: Inertia Tensor - Proof

$$L_k = (\sum_i^N m_i[r_{ix}r_{ix}\delta_{kx} - r_{ix}r_{ik}] \quad \sum_i^N m_i[r_{iy}r_{iy}\delta_{ky} - r_{iy}r_{ik}] \quad \sum_i^N m_i[r_{iz}r_{iz}\delta_{kz} - r_{iz}r_{ik}])\omega$$

Going back to L

$$\begin{pmatrix} L_x \\ L_y \\ L_z \end{pmatrix} = \begin{pmatrix} \sum_i^N m_i[r_{ix}r_{ix}\delta_{xx} - r_{ix}r_{ix}] & \sum_i^N m_i[r_{ix}r_{ix}\delta_{yx} - r_{ix}r_{iy}] & \sum_i^N m_i[r_{ix}r_{ix}\delta_{zx} - r_{ix}r_{iz}] \\ \sum_i^N m_i[r_{iy}r_{iy}\delta_{xy} - r_{iy}r_{ix}] & \sum_i^N m_i[r_{iy}r_{iy}\delta_{yy} - r_{iy}r_{iy}] & \sum_i^N m_i[r_{iy}r_{iy}\delta_{zy} - r_{iy}r_{iz}] \\ \sum_i^N m_i[r_{iz}r_{iz}\delta_{xz} - r_{iz}r_{ix}] & \sum_i^N m_i[r_{iz}r_{iz}\delta_{yz} - r_{iz}r_{iy}] & \sum_i^N m_i[r_{iz}r_{iz}\delta_{zz} - r_{iz}r_{iz}] \end{pmatrix} \omega$$

Appendix: Inertia Tensor - Proof

$$\begin{pmatrix} L_x \\ L_y \\ L_z \end{pmatrix} = \begin{pmatrix} \sum_i^N m_i[r_{ix}r_{ix}\delta_{xx} - r_{ix}r_{ix}] & \sum_i^N m_i[r_{ix}r_{ix}\delta_{yx} - r_{ix}r_{iy}] & \sum_i^N m_i[r_{ix}r_{ix}\delta_{zx} - r_{ix}r_{iz}] \\ \sum_i^N m_i[r_{iy}r_{iy}\delta_{xy} - r_{iy}r_{ix}] & \sum_i^N m_i[r_{iy}r_{iy}\delta_{yy} - r_{iy}r_{iy}] & \sum_i^N m_i[r_{iy}r_{iy}\delta_{zy} - r_{iy}r_{iz}] \\ \sum_i^N m_i[r_{iz}r_{iz}\delta_{xz} - r_{iz}r_{ix}] & \sum_i^N m_i[r_{iz}r_{iz}\delta_{yz} - r_{iz}r_{iy}] & \sum_i^N m_i[r_{iz}r_{iz}\delta_{zz} - r_{iz}r_{iz}] \end{pmatrix} \omega$$

$$\begin{pmatrix} L_x \\ L_y \\ L_z \end{pmatrix} = \begin{pmatrix} \sum_i^N m_i[r_{ix}r_{ix}\delta_{xx} - r_{ix}r_{ix}] & \sum_i^N -m_i r_{ix}r_{iy} & \sum_i^N -m_i r_{ix}r_{iz} \\ \sum_i^N -m_i r_{iy}r_{ix} & \sum_i^N m_i[r_{iy}r_{iy}\delta_{yy} - r_{iy}r_{iy}] & \sum_i^N -m_i r_{iy}r_{iz} \\ \sum_i^N -m_i r_{iz}r_{ix} & \sum_i^N -m_i r_{iz}r_{iy} & \sum_i^N m_i[r_{iz}r_{iz}\delta_{zz} - r_{iz}r_{iz}] \end{pmatrix} \omega$$

Appendix: Inertia Tensor - Proof

$$\begin{pmatrix} L_x \\ L_y \\ L_z \end{pmatrix} = \begin{pmatrix} \sum_i^N m_i[r_{ix}r_{ix}\delta_{xx} - r_{ix}r_{ix}] & \sum_i^N -m_i r_{ix} r_{iy} & \sum_i^N -m_i r_{ix} r_{iz} \\ \sum_i^N -m_i r_{iy} r_{ix} & \sum_i^N m_i[r_{iy}r_{iy}\delta_{yy} - r_{iy}r_{iy}] & \sum_i^N -m_i r_{iy} r_{iz} \\ \sum_i^N -m_i r_{iz} r_{ix} & \sum_i^N -m_i r_{iz} r_{iy} & \sum_i^N m_i[r_{iz}r_{iz}\delta_{zz} - r_{iz}r_{iz}] \end{pmatrix} \omega$$

$$L = I\omega$$

With

$$I = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} \in \mathbb{R}^{3\times 3}$$

Appendix: Torque With Inertia Tensor

- The proof for the torque $\tau = I\alpha$ is very similar but with $\dot{\omega}$ instead of ω
- You can start with

$$\tau = \mathrm{r} \times \mathrm{F}$$

The force is the force applied to every point in the rigid body, and

$$\alpha = \dot{\omega} \times r + \omega \times (\omega \times r)$$

- Hint: Look at what happens with $r \times (\omega \times (\omega \times r))$
- And is left as an exercise ;)
- (the short version is to just differentiate both sides of the previous equation, but.. Where is the fun ☺)

Appendix: Laplacian and Averages

- The laplacian is the derivative of the gradient of a field.
- Suppose you have a 1D function $f(x)$ on a grid.
- The laplacian can be approximated by

$$\nabla^2 \mathrm{f}(\mathrm{x}) \approx f(x+h) + f(x-h) - 2f(x)$$

- Where h is the space between 2 grid points

$$average = \frac{f(x+h) + f(x-h)}{2}$$

- Rearranging:

$$\nabla^2 \mathrm{f}(\mathrm{x}) \approx 2average - 2f(x)$$

- It follows:

$\nabla^2 f(x) > 0 : f(x)$ smaller than the average around it

$\nabla^2 f(x) < 0 : f(x)$ bigger than the average around it